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Coherent Ising Machines with Optical Error Correction Circuits

*Sam Reifenstein, Satoshi Kako, Farad Khoyratee, Timothée Leleu, and Yoshihisa Yamamoto**

A coherent Ising machine (CIM) comprising a network of open-dissipative quantum oscillators with optical error correction circuits is proposed. In the proposed network, the squeezed/anti-squeezed vacuum states of the constituent optical parametric oscillators establish quantum correlations below the threshold through optical mutual coupling and collective symmetry breaking is induced above the threshold as a decision-making process. This initial search process is followed by a chaotic solution search step facilitated by optical error correction feedback. The particular algorithm used by the proposed network is derived from the truncated Wigner stochastic differential equation. As an optical hardware technology, the proposed CIM has several unique features, including programmable all-to-all Ising coupling in the optical domain, directional coupling ($J_{ij} \neq J_{ji}$)-induced chaotic behavior, and the ability to operate at low power at room temperature. The proposed CIM is evaluated in terms of its performance and how this scales at different problem sizes. It is shown that the various CIMs discussed in this paper are effective at solving many problem types, although the optimal algorithm depends on the problem case. It is further shown that the proposed optical implementations can be achieved with low energy consumption on thin-film $LiNbO_3$ platforms.

1. Introduction

Combinatorial optimization problems are ubiquitous in modern science, engineering, medicine, and business. As such problems are often NP-hard, their runtime on classical digital computers is expected to scale exponentially. A representative example of an NP-hard combinatorial optimization problem is the non-planar Ising model.[1] Special-purpose quantum hardware devices have been developed for finding The solutions of Ising problems more efficiently than standard heuristic approaches. For example, quantum annealing (QA) devices exploit the adiabatic evolution of pure-state vectors using a time-dependent Hamiltonian.[2,3] Another example is the coherent Ising machine (CIM), which exploits the quantum-to-classical transition of mixed-state density operators in a quantum oscillator network.[4–7] Performance comparisons between QA devices and CIMs for various Ising models incorporating complete, dense, or sparse graphs have been reported.[8] Theoretical performance comparisons between ideal gate-model quantum computers that implement either Grover's search algorithm or the adiabatic quantum computing algorithm and CIMs have also been reported recently.[9] Although CIMs featuring all-to-all coupling among spins are highly effective, their use of external field-programmable gate array (FPGA) circuits, analog-to-digital converters (ADCs), and digital-to-analog converters (DACs) not only results in considerable energy dissipation but also introduces potential bottlenecks for high-speed operation.

The standard linear coupling scheme used by CIMs has been found to suffer from amplitude heterogeneity among the constituent quantum oscillators. Consequently, the Ising Hamiltonian is incorrectly mapped to the network loss, resulting in unsuccessful operation, particularly in frustrated spin systems.[10] A novel error-correcting feedback scheme developed to resolve this problem[11,12] increases the solution accuracy of CIMs to a level comparable to that of state-of-the-art heuristics such as break-out local search (BLS). [14] In this paper, we introduce a novel CIM architecture in which error correction is implemented optically. In the proposed architecture, computationally intensive matrix-vector multiplication (MVM) and a nonlinear feedback function are implemented using phase-sensitive (degenerate) optical parametric amplifiers, which are essentially equivalent to main-cavity optical parametric oscillators (OPOs). This new CIM architecture can potentially be implemented monolithically in future photonic integrated circuits using thin-film LiNbO3 platforms. $LiNbO_3$ platforms.[15]

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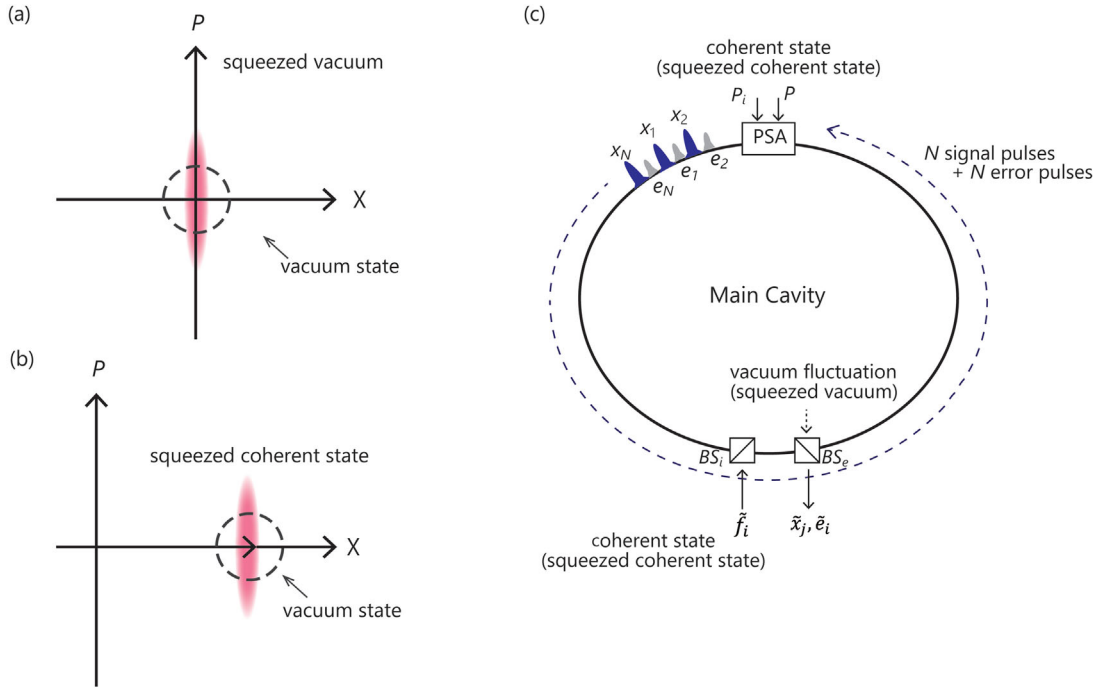

Figure 1. a) Vacuum and squeezed vacuum states. b) Coherent and squeezed coherent states. c) Conventional CIM with vacuum noise injected from reservoirs and a modified CIM with suppressed reservoir noise.

Because of their simple and efficient theoretical descriptions, networks of open dissipative quantum oscillators with optical error correction circuits are promising not only as future hardware platforms but also as tools for implementing quantum-inspired algorithms. Numerical simulation of the time evolution of an N-qubit quantum system requires 2^N complex-number amplitudes. However, various phase-space techniques in quantum optics have been developed over the last four decades to simulate quantum oscillator networks.[18–20] The complete description of a network of quantum oscillators is now possible using N (or $2N$) sets of stochastic differential equations (SDEs) based on positive-P,[21] truncated Wigner,[22–24] or truncated Husimi[23,24] representations of the master equations. These SDEs can be used as heuristic algorithms on modern digital platforms. To completely describe a network of low-Q quantum oscillators, a discrete map technique using a Gaussian quantum model, which is also computationally efficient, is available.[25]

Similarly, a network of dissipation-less quantum oscillators with adiabatic Hamiltonian modulation that can be described using a set of N deterministic equations can also be used to implement a heuristic algorithm on a modern digital platform.[27–29] Such heuristic algorithms are called simulated bifurcation machines (SBMs),[27,29] and a variant SBM will be studied in Section 6. Although the original SBM was inspired by dissipation-less adiabatic quantum computation, the version of SBM discussed in this paper (dSBM) is not a true unitary system, as dissipation is artificially added using inelastic walls to improve the performance of the algorithm. As both algorithms involve MVM as a computational bottleneck when simulated on a digital computer, we use the number of MVMs as the metric for performance comparison. We find that both types of systems have very similar performance in most cases except on graph types with significant variation in vertex degree, where the SBM struggles consistently.

2. Semi-Classical Model for Error Correction Feedback

In this section, we will describe several mutual coupling and error correction feedback schemes for CIMs. To simplify our argument, we consider a semi-classical deterministic picture.[10] The semi-classical model treated in this section is an approximate theory for the following fictitious machine. At an initial time $t = 0$, each signal pulse field is prepared in a vacuum state (**Figure 1**a) and each error pulse field is prepared in a weak coherent state (Figure 1b). When the pump fields p and p_i are switched on at $t \geq 0$, a vacuum field incident on the extraction beam splitter BS_e from an open port is squeezed/anti-squeezed by a phase-sensitive amplifier (PSA) in the optical delay line (ODL) CIM, as shown in Figure 1c. In other words, the vacuum fluctuation in the in-phase component $\tilde{X} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger)$ is deamplified by a factor of $1/G$ whereas the vacuum fluctuation in the quadrature-phase component $\hat{P} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger)$ is amplified by a factor of G. Similarly, the vacuum fluctuations incident on the OPO pulse field arising from linear loss in the cavity are all squeezed by the respective PSA. The pump and feedback injection field fluctuations along the in-phase component are also deamplified by the respective PSA (Figure 1c).

The truncated Wigner stochastic differential equation (W-SDE) for such a quantum-optic CIM with squeezed reservoirs has been derived and studied previously.[31] This particular CIM achieves maximum quantum correlation among OPO pulse

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fields along the in-phase component and maximum success probability[31] because, in such systems, the quantum correlation among the OPO pulse fields is formed by the mutual coupling of the vacuum fluctuations of these fields without the injection of uncorrelated fresh reservoir noise. The following semi-classical model is considered as an approximate theory of the W-SDE described above in the limit of a large deamplification factor ($G \gg 1$); a full quantum description of a more realistic CIM with optical error correction circuits (without reservoir engineering) is given in Section 5.

To overcome the problem of amplitude heterogeneity in the CIM,[10] the addition of an auxiliary variable for error detection and correction has been proposed.[11,12] This system has been studied as a modification of the measurement feedback CIM.[31] The spin variable (signal pulse amplitude) x_i and auxiliary variable (error pulse amplitude) e_i obey the following deterministic equations[11]

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = -x_i^3 + (p-1)x_i - e_i \sum_j \xi J_{ij} x_j \tag{1}$$

$$\frac{\mathrm{d}e_i}{\mathrm{d}t} = -\beta e_i \left(x_i^2 - \alpha\right) \tag{2}$$

where J_{ij} is the Ising coupling matrix, α, β, and p are system parameters, and ξ is a normalizing constant for J_{ij} (see Appendix A for parameter selection). In many cases, these parameters can be modulated over time to achieve better performance (see Section 3 and Appendix C). To use this system as an Ising solver, we consider the spin configuration $\sigma_i = \mathrm{sign}(x_i)$ as a possible solution to the corresponding Ising problem. Even though noise is ignored in Equations (1) and (2), the initial amplitude x_i can be chosen randomly to create a diverse set of possible trajectories.

In this paper, we refer to this system of equations as CIM with chaotic amplitude control (CIM-CAC). The term "chaotic" is used because CIM-CAC exhibits chaotic behavior (as discussed in Section 3). CIM-CAC can refer to either the above-described system of deterministic differential equations when integrated as a digital algorithm or an optical CIM that emulates the corresponding equations.

To study the CIM-CAC equations, we have made the following modifications

$$z_i = e_i \sum_j \xi J_{ij} x_j \tag{3}$$

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = -x_i^3 + (p-1)x_i - z_i \tag{4}$$

$$\frac{\mathrm{d}e_i}{\mathrm{d}t} = -\beta e_i \left(z_i^2 - \alpha\right) \tag{5}$$

which we refer to as CIM with chaotic feedback control (CIM-CFC). The only difference between Equations (2) and (5) is that, in the latter, the time evolution of the error variable e_i monitors the feedback signal z_i rather than the internal amplitude x_i. The dynamics of this new equation are very similar to those of CIM-CAC, as highlighted by the observation that CIM-CAC and CIM-CFC have nearly identical fixed points. Our motivation for studying CIM-CFC in addition to CIM-CAC is to gain a better understanding of how these systems work. In addition, CIM-CFC can have slightly simpler dynamics, which in turn simplifies its numerical integration.

The third system discussed in this paper has a very different equation

$$z_i = \sum_j \xi J_{ij} x_j \tag{6}$$

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = -x_i^3 + (p-1)x_i - f\left(cz_i\right) - k\left(z_i - e_i\right) \tag{7}$$

$$\frac{\mathrm{d}e_i}{\mathrm{d}t} = -\beta\left(e_i - z_i\right) \tag{8}$$

where the non-linear function f is a sigmoid-like function such as $f(z) = \tanh(z)$ and p, k, c, and β are system parameters (see Appendix A for parameter selection). The significance of this new feedback system is that the differential equation for the error signal e_i is now linear in the "mutual coupling signal" z_i. In addition, z_i is calculated simply as $\sum_j \xi J_{ij} x_j$ without the additional factor e_i used in Equation (6). This means that the only nonlinear elements in the system are the gain saturation term $-x_i^3$ and the nonlinear function f. To obtain the results reported in this paper, we used $f(z) = \tanh(z)$; however, if a different function with the same properties is used, the system will have similar behavior.

In the system described above, the two essential aspects of CIM-CAC and CIM-CFC are separated into two different terms. The term $f(cz_i)$ produces mutual coupling while passively addressing the problem of amplitude heterogeneity; the term $k(z_i - e_i)$ introduces the error signal e_i, which helps to destabilize local minima. Therefore, we refer to this system as CIM with separated feedback control (CIM-SFC) in the remainder of this paper.

Another significant aspect differentiating CIM-SFC (Equations (6)–(8)) from CIM-CAC and CIM-CFC (Equations (1)–(5)) is that the auxiliary variables e_i in CIM-SFC have very different meanings. In CIM-CAC and CIM-CFC, e_i is meant to be a strictly positive number that varies exponentially and modulates the mutual coupling term. In CIM-SFC, by contrast, e_i is a variable that stores sign information and takes the same range of values as the mutual coupling signal z_i. The error signal e_i and the term $k(e_i - z_i)$ can essentially be regarded, respectively, as low- and high-pass filters on z_i (in other words, $k(e_i - z_i)$ only registers sharp changes in z_i). The similarities and differences among CIM-SFC, CIM-CAC, and CIM-CFC can be ascertained by observing the fixed points. In CIM-CAC and CIM-CFC, the fixed points are of the form[11]

$$x_i = \lambda_1 \sigma_i \tag{9}$$

$$e_i = \lambda_2 \frac{1}{h_i \sigma_i} \tag{10}$$

with

$$h_i = \sum_j \xi J_{ij} \sigma_j \tag{11}$$

where σ_i is a spin configuration corresponding to a local minimum of the Ising Hamiltonian and λ_1 and λ_2 are constants that depend on the system parameters. In CIM-SFC, the fixed points

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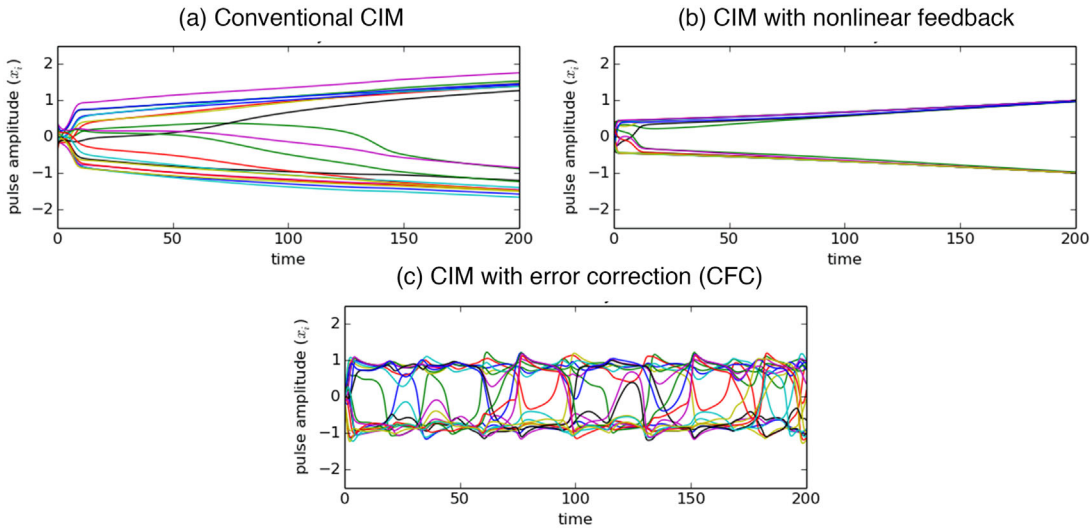

Figure 2. Trajectories of OPO amplitudes in CIMs with a) linear feedback, b) nonlinear filtering feedback, and c) chaotic feedback control.

are generally very complicated and difficult to express explicitly. However, if we consider the limit of $c \gg 1$, the fixed points will take the form

$$x_i = \lambda \sigma_i \tag{12}$$

$$e_i = \lambda h_i \tag{13}$$

where λ is a number such that $-\lambda^3 + (p-1)\lambda = -1$ and σ_i is again a spin configuration corresponding to a local minimum. This formula is only valid if $f(cz)$ is an odd function that takes the value of $+1$ for $cz \gg 1$ and -1 for $cz \ll -1$. Therefore, it is important to choose an appropriate function f.

The important difference between the fixed points of these two types of systems is that, in CIM-CAC and CIM-CFC, the error signal is

$$|e_i| \propto \frac{1}{|h_i|} \tag{14}$$

whereas in CIM-SFC, the error signal is

$$|e_i| \propto |h_i| \tag{15}$$

In Section 5, we will see that this difference makes CIM-SFC more robust to quantum noise from reservoirs and pump sources. In the next section, we will investigate the similarities and differences among these three systems using numerical simulation.

3. Numerical Simulation of CIM-CAC, CIM-CFC, and CIM-SFC

The originally proposed CIM architecture employs simple linear feedback without an error detection/correction mechanism. In other words, the feedback term in Equation (1) is simply $\sum_j \xi J_{ij} x_j$.[10] In this case, the Ising Hamiltonian cannot be properly mapped to the network loss owing to OPO amplitude heterogeneity, particularly in frustrated spin systems, as shown in **Figure 2**a. Such a CIM often does not find a ground state of the Ising Hamiltonian; instead, it selects the lowest-energy eigenstate of the coupling (Jacobian) matrix $[J_{ij}]$.[10] This undesired operation is caused by a system's formation of heterogeneous amplitudes.[10] This problem can be partially addressed by introducing a nonlinear filter function such as $\tanh(\sum_j \xi J_{ij} x_j)$ for the feedback pulse to achieve homogeneous OPO amplitudes—at least well above threshold—as shown in Figure 2b and satisfy the proper mapping condition toward the end of the system's trajectory. However, such nonlinear filtering alone is not sufficiently powerful to prevent the machine from being trapped in numerous local minima. As the problem size N increases, NP-hard Ising problems are expected to have exponentially increasing numbers of local minima; thus, a system that is easily trapped in these minima will be ineffective.

To destabilize the attractors caused by local minima and allow the machine to continue searching for a true ground state, the error detection/correction variable expressed by Equation (2) or (5) can be introduced. As shown in Figure 2c, the trajectory of a CIM with error correction variables will not reach equilibrium but continue to explore many states. Conversely, the systems in Figure 2a,b, which do not have the error correction variable (e_i), will often converge rapidly on a fixed point corresponding to a high-energy excited state of the Ising Hamiltonian. Destabilizing the local minima will inevitably make the ground state unstable as well. Although this is undesirable, the machine can be simply allowed to visit many local minima and subsequently determine which has the lowest energy. Alternatively, we have found that modulating the system parameters can give the system a high probability of staying in a ground state toward the end of its trajectory (see Section 4 for further details).

The addition of another N degrees of freedom allows the machine to visit a local minimum, escape from it, and continue to explore nearby states; this is not possible with a conventional CIM algorithm. In this section, we will discuss the dynamics of the error correction schemes, CIM-CFC and CIM-SFC, proposed in this paper.

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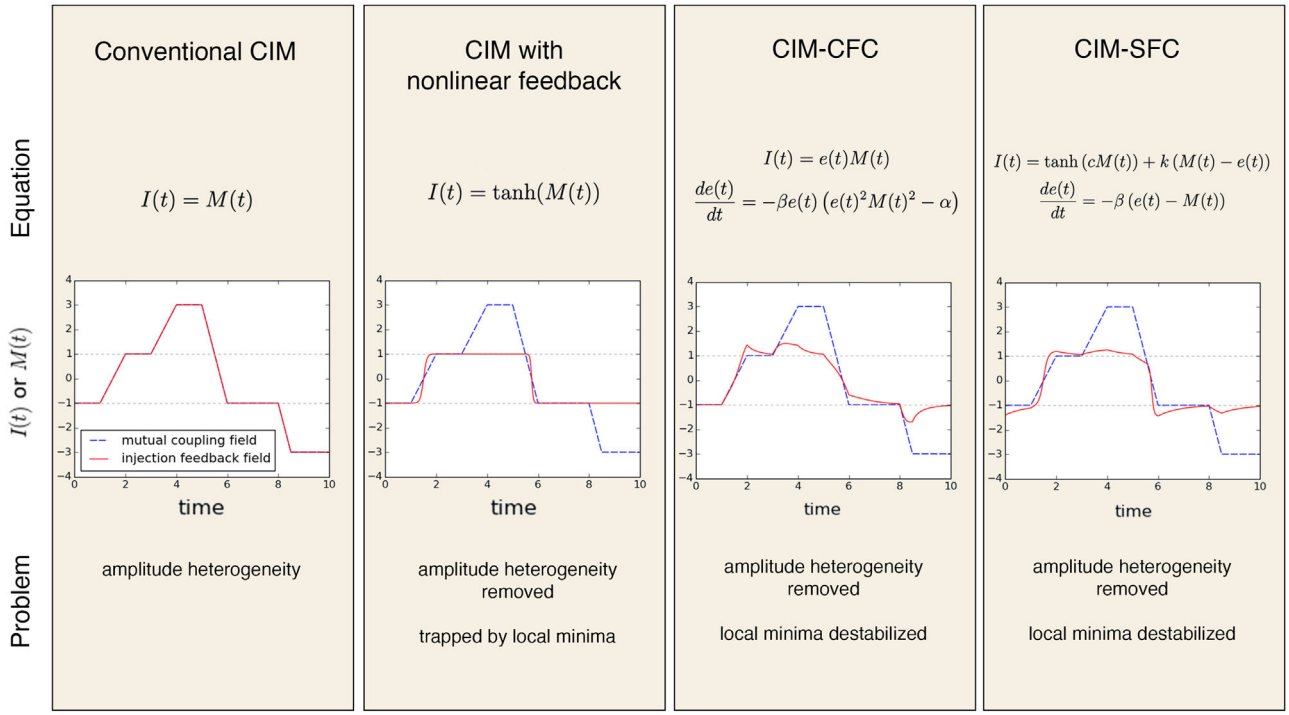

Figure 3. Mutual coupling field (blue) and injection feedback field (red) in four different feedback systems.

Even though CIM-CFC and CIM-SFC are described by very different equations, the two systems are derived from a similar concept. To understand why CIM-CFC and CIM-SFC are similar, we can consider systems in which the "mutual coupling signal" $M_i(t) = \sum_j \xi J_{ij} x_j(t)$ and the "injection feedback signal" $I_i(t)$ are introduced. We can then write both CIM-CFC and CIM-SFC in the form

$$M_i(t) = \sum_j \xi J_{ij} x_j(t) \tag{16}$$

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = -x_i^3 + (p-1)x_i - I_i(t) \tag{17}$$

where $I_i(t)$ depends on the time evolution of $M_i(t)$. **Figure 3** shows how $I_i(t)$ (red) varies with respect to a mutual coupling field $M_i(t)$ (blue) for four different feedback schemes.

CIM-CFC and CIM-SFC share the following similarities under these conditions. If the mutual coupling field $M_i(t)$ remains constant for a certain period of time, the injection feedback field $I_i(t)$ will converge on the value given by $\mathrm{sign}(M_i(t))$. If, however, $M_i(t)$ varies sharply, $I_i(t)$ will deviate from its steady state values $+1/-1$. This small deviation is effective at triggering destabilization when the system is near a local minimum, which allows the machine to explore new spin configurations.

Although CIM-CFC and CIM-SFC are derived from the same principle, the dynamics of the two systems appear to differ from each other. In particular, CIM-CFC (and CIM-CAC) nearly always features chaotic dynamics, as the trajectory is highly sensitive to the initial conditions. In the case of CIM-SFC, the trajectory will often immediately fall into a stable periodic orbit unless the parameters are dynamically modulated. At present, we do not have an exact theoretical reason for this difference in dynamics; this is purely an experimental observation. A more theoretical analysis of the case of CIM-CAC can be found elsewhere.[11]

To demonstrate this difference, **Figure 4** shows the correlation of pulse amplitudes between two initial conditions that are very close to each other. The initial condition for the pulse amplitude #1 (plotted on the x-axis) is chosen from a zero-mean Gaussian with a standard deviation of 0.25; the initial condition for trajectory #2 (plotted in y-axis) is equal to that of trajectory #1 plus a small amount of noise (standard deviation 0.01).

Figure 4 shows the correlation of all 100 pulse amplitudes between the two initial conditions for a Sherrington–Kirkpatrick (SK) spin glass instance of problem size $N = 100$. In CIM-SFC (first row), the correlation remains even after 4000 time-steps (round trips), indicating that the two initial conditions follow a nearly identical trajectory. However, in CIM-CFC (second row), the x_i variables become uncorrelated after around 100 time-steps even though the initial conditions of the two trajectories are very close. This indicates qualitatively that CIM-CFC is highly sensitive to the initial condition whereas CIM-SFC is not.

This pattern tends to hold when different parameters and initial conditions are used. However, although CIM-SFC stays correlated in most cases, the two trajectories diverge under certain system parameters and initial conditions. This means that, although CIM-SFC is less sensitive to the initial conditions than CIM-CFC, some chaotic dynamics likely occur during the search, especially when the parameters are modulated.

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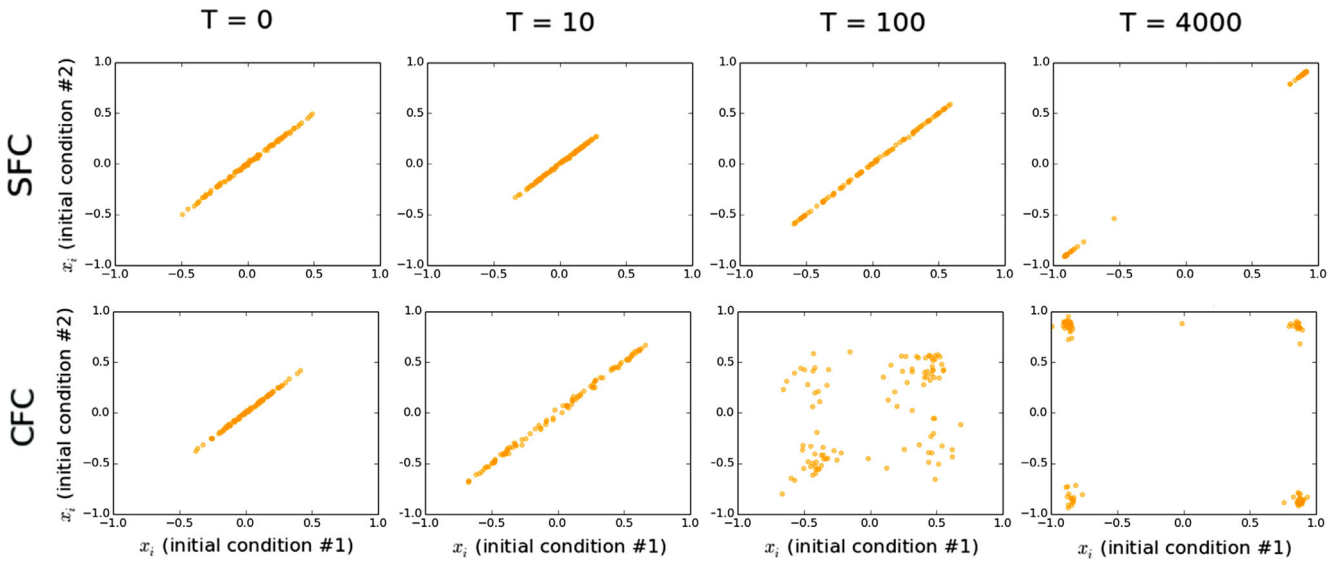

Figure 4. Signal pulse amplitude correlations at different evolution times in CIM-SFC and CIM-CFC.

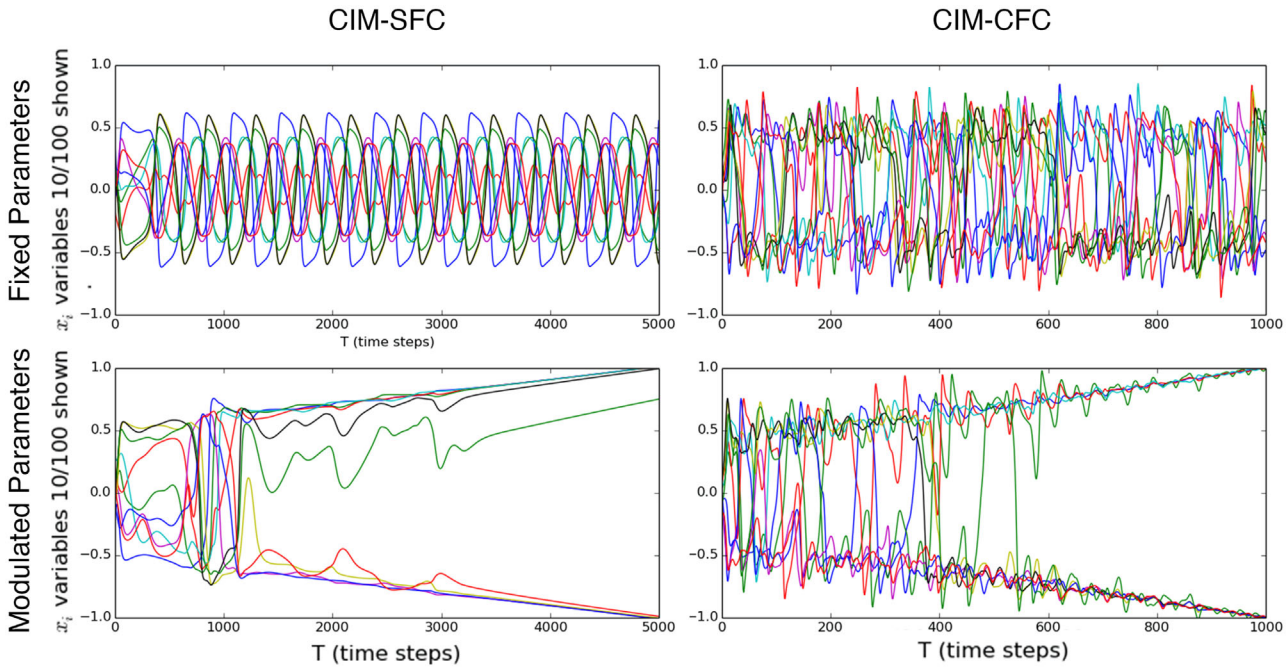

Figure 5. Signal pulse amplitude trajectories of CIM-SFC and CIM-CFC with fixed and modulated system parameters.

Another way to qualitatively observe the difference in dynamics is to simply observe the trajectories. **Figure 5** shows examples of trajectories of both systems (10 out of 100 x_i variables are shown) with fixed and linearly modulated system parameters. When the parameters are fixed, the difference between the two systems is evident: CIM-SFC will rapidly become trapped in a stable periodic attractor, whereas CIM-CFC will continue to search in an unpredictable manner, allowing the parameters to be slowly modulated so that the system can find a ground state. Although CIM-CFC and CIM-CAC can find ground states with fixed parameters, we have found that modulation of the system parameters improves the performance of both considerably (see Appendix C for details).

In the lower left panel of Figure 5, the parameters c and p of CIM-SFC increase linearly from low to high values (p ranges from -1 to $+1$ and c ranges from 1 to 3). We see that, as the parameters change, the system can jump from one attractor to another and eventually end up in a fixed point/local minimum. By linearly increasing the parameters c and p in CIM-SFC from low to high values, we slowly transition the nonlinear term $\tanh(cz_i)$

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from a "soft spin" mode in which the nonlinear coupling term has a continuous range of values between −1 and 1 to a "discrete" mode in which $\tanh(cz_i)$ primarily assumes the values +1 or −1. This transition appears to be crucial for CIM-SFC to function properly.

For most fixed parameters, CIM-SFC rapidly approaches a periodic or fixed-point attractor, as shown in Figure 5; however, as mentioned earlier, it is likely that for some specific values of c and p CIM-SFC will feature chaotic dynamics similar to those of CIM-CFC. It has been shown[11,13] that chaotic dynamics are observed when solving hard optimization problems efficiently using a deterministic system. This trend is also observed in the SBM.[27,29] Whether or not CIM-SFC utilizes chaotic dynamics, however, is beyond the scope of this paper; to answer this question, we would need to further analyze how the parameters of CIM-SFC affect its dynamics and gain a deeper understanding of how CIM-SFC finds ground states.

4. Implementation of CIM with Optical Error Correction Circuits

Figure 6, together with 1c, show the physical setup of a CIM-CAC or CIM-CFC system with optical error correction circuits. In our design, the main ring cavity stores signal pulses with normalized amplitude, x_i, and error pulses with normalized amplitude, e_i, where $i = 1, 2, \ldots, N$. The signal pulses start from vacuum states $|0\rangle_1|0\rangle_2 \cdots |0\rangle_N$ and are amplified (or deamplified) along the X-coordinate by a positive (or negative) pump rate p.

The error pulses start from a coherent state $|\alpha\rangle_1|\alpha\rangle_2 \cdots |\alpha\rangle_N$ with $\alpha > 0$ and are amplified (or deamplified) along the X-coordinate by the pump rate p' as described below. The squared amplitudes of the error pulses are kept small ($e_i^2 < 1$) relative to the saturation level of the main cavity OPO; thus, the error pulses are controlled within a linear amplifier/deamplifier regime whereas the signal pulses are controlled within both a linear amplifier/deamplifier ($x_i^2 < 1$) and a nonlinear oscillator regime ($x_i^2 > 1$).

The extraction beamsplitter BS_e in Figure 1c selects partial waves of the signal and error pulses that are amplified by the noise-free phase-sensitive amplifier PSA_0, as shown in Figure 6a). PSA_0 amplifies the signal and error pulses to a classical level without introducing additional noise. Although the extracted amplitudes $\tilde{x}_i$ and $\tilde{e}_i$ suffer from signal-to-noise ratio (SNR) degradation owing to the vacuum noise incident on BS_e, they are amplified by the high-gain noise-free PSA_0 to classical levels; therefore, no further SNR degradation occurs despite the large linear losses in the optical error correction circuits.

A small part of the output of PSA_0 is sent to an optical homodyne detector that measures the extracted signal and error pulses, which have amplitudes $\tilde{x}_i$ and $\tilde{e}_i$, respectively. The measurement error of the homodyne detection is determined solely by the reflectivity of BS_e and the vacuum fluctuation incident on BS_e (as described above). Figure 6a shows the output of the fan-out circuit at different times $t = \tau, 3\tau, 5\tau, \ldots$ separated by signal-pulse-to-signal-pulse intervals of 2τ.

For instance, the signal pulse ($\tilde{x}_j$) is first input to PSA_j and then sent to optical delay line DL_j with a delay time of $(2N - 2j + 1)\tau$. The phase-sensitive gain/loss of PSA_j is set to $\sqrt{G_j} = \xi J_{ij}$ so that the amplified/deamplified signal pulse that arrives in front of the fan-in circuit at time $t = 2N\tau$ is equal to $\xi J_{ij}\tilde{x}_j$. Therefore, the fan-in circuit will output a pulse with the desired amplitude of $\sum_j \xi J_{ij}\tilde{x}_j$. If PSA_j has a phase-sensitive linear gain/loss of 10dB, we can implement an arbitrary Ising coupling of range $10^{-2} < |\xi J_{ij}| < 1$.

The output of the fan-in circuit is then input to another PSA, PSA_e, that amplifies with a factor of $\sqrt{G_e} = \tilde{e}_i$. This is achieved by modulating the pump power to PSA_e based on the measurement result for $\tilde{e}_i$. Finally, the output of PSA_e is injected back into signal pulse (x_i) in the main cavity via BS_i (see Figure 1c). The BS_e outputs not only signal pulses but also error pulses that are used only for homodyne detection. Thus, the pump power to PSA_0 is switched off for the error pulses to deamplify the residual error pulses by PSA_1, PSA_2, ..., PSA_N, PSA_e. In this manner, spurious injection of error pulses back into the main cavity is avoided, and the error pulse dynamics are governed solely by the pump power p_i to the main cavity PSA, which is set to satisfy $p_i - 1 = \beta(\alpha - \tilde{x}_i^2)$ or $p_i - 1 = \beta(\alpha - \tilde{z}_i^2)$.

One advantage of this optical implementation of CIM-CAC and CIM-CFC is that only one type of active device—a noise-free phase-sensitive (degenerate optical parametric) amplifier (PSA)—is needed, and all other elements are passive devices. This allows for the potential on-chip monolithic integration of the CIM system as well as low-energy dissipation in the computational unit, which will be discussed in Section 5.

A similar optical implementation of CIM-SFC is shown in Appendix F.

Figure 6b shows the pump pulse factory that provides the second harmonic generation (SHG) pulses to the main cavity PSA, post-amplifier PSA_0, delay line amplifiers PSA_1, PSA_2, $\cdots$ PSA_N, and exit amplifier PSA_e. The purpose of this pump pulse factory is to reduce the use of EOM modulators, which consume the bulk of the energy within the overall CIM. A soliton frequency comb generator produces a pulse train at a repetition frequency of 100 GHz and wavelength of 1.56 μm. Before it is split into multiple branches, the pulse train is amplified by a pump amplifier PSA_p. N storage ring cavities continuously produce pump pulses for PSA_1, PSA_2, $\cdot$ PSA_N in order to implement the MVM $\sum \xi J_{ij}\tilde{x}_j$. For this purpose, the pulses stored in the ith ring cavity are amplified to the appropriate amplitudes to achieve a gain of $\sqrt{G_{ij}} = \xi J_{ij}$. The time needed to use N EOM arrays is only one round trip of the ring cavity, that is, $N \times 10$ (ps). The out-coupling loss of the storage ring cavities is compensated by the linear gain of the internal PSA_s. The pump pulses for PSA_p, PSA_s, and PSA_0 have constant amplitudes and are therefore driven directly by the PSA_p output. The pump pulses P and P_i for the signal and error pulses in the main cavity, as well as the exit PSA_e, must be modulated over the entire computation time.

Another factor that must be accounted for in an optical implementation is the calculation of the Ising energy. In our digital simulation for generating the results presented in this paper, the Ising energy was calculated at each time step (round trip) and the smallest energy obtained was used as the result of the computation. This means that, in an optical implementation, it is necessary to measure the $\tilde{x}_i$ amplitude at every round trip and calculate the Ising energy using, for instance, an external

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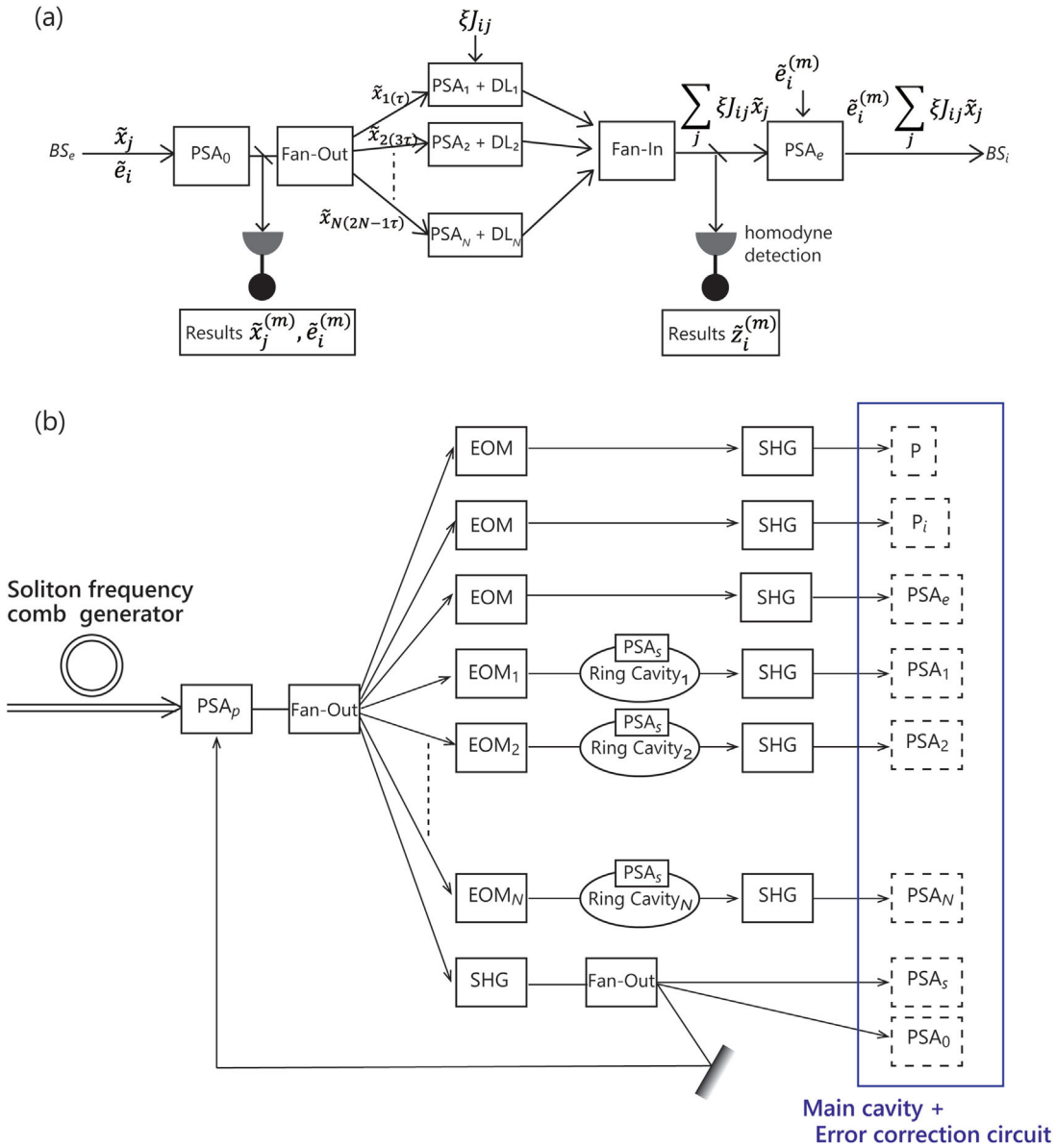

Figure 6. a) Optical implementation of error correction circuits for CIM-CAC and CIM-CFC. b) Pump pulse factory providing SHG pulses to the main cavity, and error correction circuits. The pump pulse factory carries N^2 pulses spread over N optical cavities corresponding to the elements of the Ising coupling matrix J_{ij}.

ADC/FPGA circuit. This would defeat the purpose of using optics, as the digital circuit in the ADC/FPGA would become a bottleneck in terms of time and energy consumption.

However, we have found that, by applying proper parameter modulation as shown in Figure 5, it is possible to use only the final state of the system for the result and still have a high probability of success. To obtain the results for 800-spin Ising instances (SK model) presented in Section 6, we calculated how often a successful trajectory was in the ground spin configuration after the final time step. For CIM-SFC, we found that in 100% of the 7401 successful trajectories the final spin configuration were in the ground state. In other words, if CIM-SFC visits the ground state at any point during the trajectory, it will also be in the ground state at the end of the trajectory. For CIM-CFC and CIM-CAC, this was true only 75% and 48% of the time, respectively. We believe that this difference among the three systems is a result of both the intrinsic dynamics and the parameters used.

This suggests that, for a CIM with optical error correction, it is possible to simply digitize the final measured value of x_i obtained after many round trips to obtain a computational result and still have a high probability of success. In the case of CIM-CFC and CIM-CAC, it might be beneficial to read the spin configuration multiple times during the last few round trips, as the machine usually visits the ground state close to the end of the trajectory even if it does not stay there.

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5. Quantum Noise Analysis and Energy Cost to Solution

As the proposed dynamical systems are to be implemented on analog optical devices, it is important to investigate the extent to which the noise from the physical system (in this case, quantum noise from pump sources and external reservoirs) will degrade their performance. In this section, we present quantum models based on our optical implementation.

In our optical implementation for CIM-CAC, the real-number signal pulse amplitude μ_i (in units of photon amplitude) obeys the following truncated W-SDE[22,23]

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu_i = (p-1)\mu_i - g^2\mu_i^3 + \tilde{\nu}_i \sum_j \xi J_{ij}\tilde{\mu}_j + n_i \tag{18}$$

where $p\mu_i$ and $-\mu_i$ represent the parametric linear gain and the linear loss rate, respectively; the latter term includes the cavity background and extraction/injection beam splitter losses for mutual coupling and error correction. The nonlinear term $-g^2\mu_i^3$ represents gain saturation (or back-conversion from signal to pump), where g is the saturation parameter. The saturation photon number is given by $1/g^2$, which is equal to the average photon number of a solitary OPO at a pump rate of $p = 2$ (two times above the threshold). Furthermore, J_{ij} is the (i,j)th element of the $N \times N$ Ising coupling matrix, as described in Section 2. As the time t is normalized by a linear loss rate, the signal amplitude decays by a factor of $1/e$ at time $t = 1$. In addition, $\tilde{\mu}_j = \mu_j + \Delta\mu_j$ and $\tilde{\nu}_i = \nu_i + \Delta\nu_i$ are the inferred amplitudes for the signal and error pulses, respectively, and $\Delta\mu_j$ and $\Delta\nu_i$ represent the additional noise terms governed by vacuum fluctuations incident on the extraction beam splitter, which are characterized by $\sqrt{\frac{1-R_\mathrm{B}}{4R_\mathrm{B}}}w$, where R_B is the reflectivity of the extraction beam splitter and w is a zero-mean Gaussian random variable with a variance of one. Finally, n_i is the noise injected from external reservoirs and pump sources[22,23]; it is characterized by the two time correlation functions $\langle n_i(t)n_i(t')\rangle = (\frac{1}{2} + g^2\mu_i^2)\delta(t-t')$. We assume that the external reservoirs are in vacuum states and that the pump fields are in coherent states.

The real number error pulse amplitude ν_i (in units of photon amplitude) is governed by

$$\frac{\mathrm{d}}{\mathrm{d}t}\nu_i = \left(p_i' - 1\right)\nu_i + m_i \tag{19}$$

where the correlation function for the noise term is given by $\langle m_i(t)m_i(t')\rangle = \frac{1}{2}\delta(t-t')$. The pump rate p_i' for the error pulse is determined by the inferred signal pulse amplitude $\tilde{x}_i = g\tilde{\mu}_i$ normalized by the saturation parameter

$$p_i' - 1 = \beta\left(\alpha - \tilde{x}_i^2\right) \tag{20}$$

The error pulses start from coherent states $|\gamma\rangle_1|\gamma\rangle_2 \cdots |\gamma\rangle_N$, for some positive real number $1/g \gg \gamma > 0$. The absence of a gain saturation term in Equation (19) implies that the error pulses are always pumped at below the threshold. Nevertheless, the error pulses represent exponentially varying amplitudes.

The parameter β governs the time constant for the error correction dynamics, and α is the squared target amplitude. This feedback model stabilizes the squared signal pulse amplitude $\tilde{x}_i^2 = g^2\tilde{\mu}_i^2$ to α through an exponentially varying error pulse amplitude $e_i = g\nu_i$. Equations (18) and (19) can be rewritten for the normalized amplitudes x_i and e_i as

$$\frac{\mathrm{d}}{\mathrm{d}t}x_i = (p-1)x_i - x_i^3 + \tilde{e}_i \sum_j \xi J_{ij}\tilde{x}_j + gn_i \tag{21}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}e_i = \left(p_i' - 1\right)e_i + gm_i \tag{22}$$

Except for the noise terms, these formulations are identical to Equations (1) and (2).

The experimental setup shown in Figure 6 can also be used to produce a CIM-CFC. In this case, the relevant truncated W-SDE for the error pulse amplitude is still given by Equation (19) or (22) but the pump rate p' should be modified to

$$p_i' - 1 = \beta\left(\alpha - \tilde{z}_i^2\right) \tag{23}$$

with

$$\tilde{z}_i = \sum_j \xi J_{ij}\tilde{x}_j \tag{24}$$

Finally, a CIM-SFC can also be produced using the experimental setup shown in Appendix F (Figure F1). In this case, Equations (21) and (22) should be modified to

$$\tilde{z}_i = \sum_j \xi J_{ij}\tilde{x}_j \tag{25}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}x_i = (p-1)x_i - x_i^3 + k\left(\tilde{e}_i - \tilde{z}_i\right) + \tanh\left(c\tilde{z}_i\right) + gn_i \tag{26}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}e_i = -\beta\left(e_i - \tilde{z}_i\right) + gm_i \tag{27}$$

A comparison of the semi-classical nonlinear dynamical models of the CIM-CAC, CIM-CFC, and CIM-SFC represented by Equations (1)–(8) with the quantum nonlinear dynamical models (truncated Wigner SDE) represented by Equations (21)–(27) reveals that they differ primarily in terms of the absence or presence of the vacuum and pump noise terms gn_i and gm_i, respectively. The other important difference is that $\tilde{x}_i$ and $\tilde{e}_i$ are inferred amplitudes of the vacuum noise contribution in the quantum model, whereas in the semi-classical model the amplitudes x_i and e_i can be reproduced without additional noise.

We next discuss the impact of quantum noise on the performance of a CIM. As indicated in Equations (21)–(27), the relative magnitude of the quantum noise in the signal and error pulses is governed by the saturation parameter g; when g increases, the ratio between the normalized pulse (x_i, e_i) and normalized quantum noise amplitudes (gn_i, gm_i) decreases. Therefore, CIM performance is expected to degrade as g increases. However, the OPO threshold pump power decreases as g increases, (see Figure C1 in ref. [31]), which suggests that the OPO energy cost to solution can be potentially reduced by increasing g.

Figure 7 shows the probability of success P_s for $N = 100$ Ising problems (SK model) plotted against the saturation parameter g^2. The reflectivity of the extraction beam splitter R_B is assumed to be $R_\mathrm{B} = 0.1$. P_s is practically independent of g^2 as long as $g^2 \lesssim 10^{-4}$.

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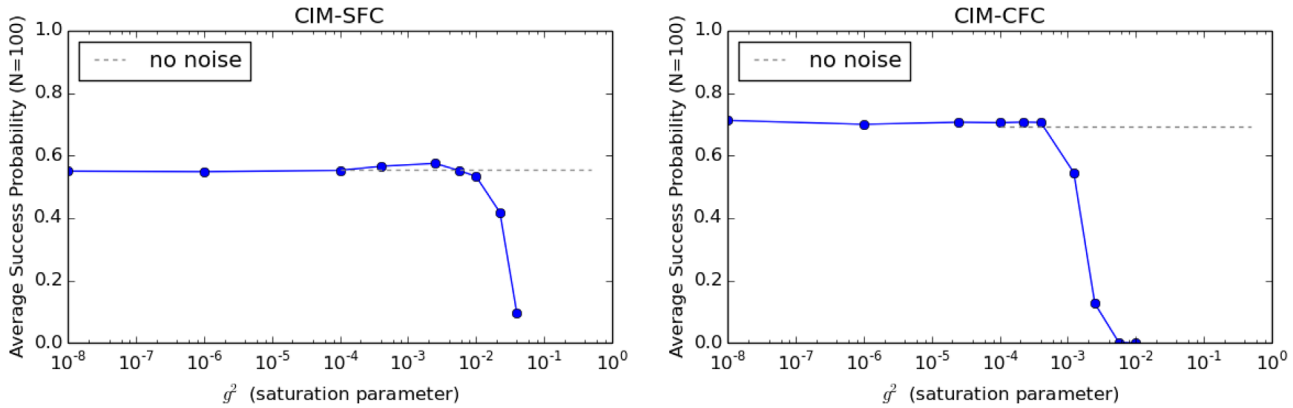

Figure 7. Success probability P_s versus saturation parameter g^2 for CIM-SFC and CIM-CFC at $N = 100$. The success probability is averaged over 100 SK instances. CIM-CAC is not shown; however, the result is nearly identical to that for CIM-CFC.

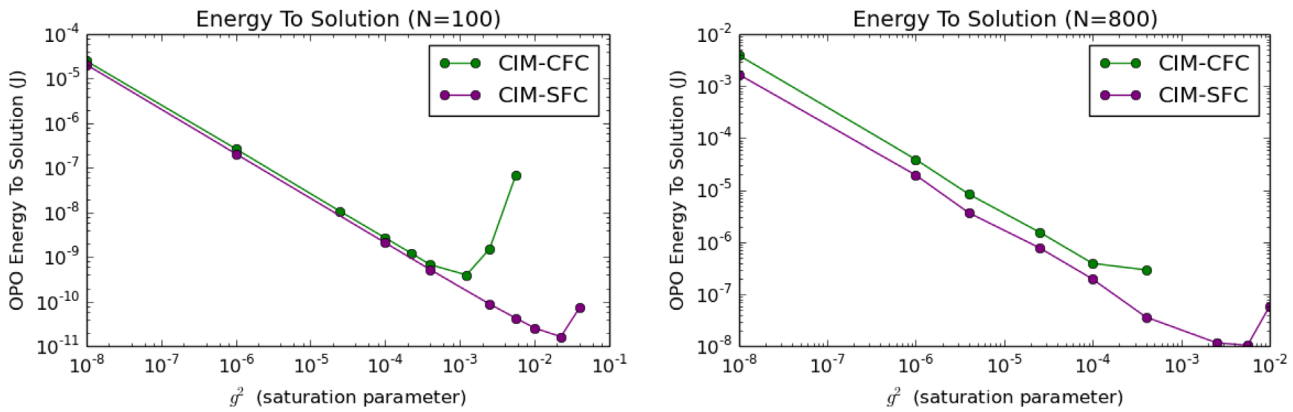

Figure 8. Energy cost to solution in joules of CIM-SFC and CIM-CFC considering only pump power to main cavity PSA. The median ETS is plotted as a function of g^2 for $N = 100$ and $N = 800$ SK instances to show the optimal value of g^2 in each case.

However, when g^2 exceeds 10^{-3}, the probability of success drops rapidly owing to the decreased signal-to-quantum noise ratio, as mentioned above.

Figure 8 shows the energy cost to solution for Ising problems (SK model) with $N = 100$ and $N = 800$ with only the pump power to the main cavity PSA, $E_{\text{main}} = 2\hbar\omega(\text{MVM})N\Delta t/g^2$, considered, where MVM is the number of matrix-vector multiplication steps to solution and Δt is a round-trip time normalized by the signal lifetime (≈ 0.1).

It is seen from the figure that CIM-SFC is more robust than CIM-CFC to quantum noise, allowing for the use of a potentially larger value of g^2. This is an expected result of the different roles played by the error variable e_i in the respective systems. In CIM-CFC, the feedback signal is calculated as

$$\tilde{z}_i = \tilde{e}_i \sum_j \xi J_{ij} \tilde{x}_j(t) \tag{28}$$

and is the primary cause of performance degradation when the quantum noise is increased; this is because a large coherent excitation of $\tilde{e}_i$ will amplify small errors in $\sum_j J_{ij}\xi\tilde{x}_j(t)$ and, conversely, large coherent excitation of $\sum_j \xi J_{ij}\tilde{x}_j(t)$ will amplify small errors in $\tilde{e}_i$. There are no such beat noise components in CIM-SFC, making it more robust to quantum noise. Moreover, the nonlinear function $\tanh(c\tilde{z}_i)$ can help to suppress the quantum noise.

Although they are not shown, the results for CIM-CAC are nearly identical to those for CIM-CFC.

If the energy cost in the optical error correction circuit and pump pulse factory (as described in Figure 6) are included, the energy cost is increased by several orders of magnitude, as shown in **Figure 9**. Here, we assume that the pump pulse energy for a small signal amplification of (≈10 dB) in PSA_1, PSA_2, ..., PSA_N, and PSA_e in the optical error correction circuit is 100 fJ per pulse and that for a large signal amplification of (≈50 dB) in PSA_0 is 1 pJ per pulse. These numbers correspond to the experimental values for a thin-film LiNO_3 ridge waveguide diode-pumped OPO (DOPO) at a pump wavelength of 780 nm and a pump pulse duration of 100 fs.[15] The pump energy consumed in the optical error correction circuit is estimated to be $E_{\text{correction}} = [(N+1)\times 10^{-13} + 10^{-12}]N(\text{MVM})(J)$. The energy consumption of the pump pulse factory can be attributed to three components: the 100-GHz soliton frequency comb generator, EOM modulators, and phase-sensitive amplifiers (Figure 6b). The 100-GHz soliton frequency comb generator requires an input power of ≈100 mW.[16] The 100-GHz EOM modulators require an electrical input power of ≈400 mW each.[17] The energy cost per pulse

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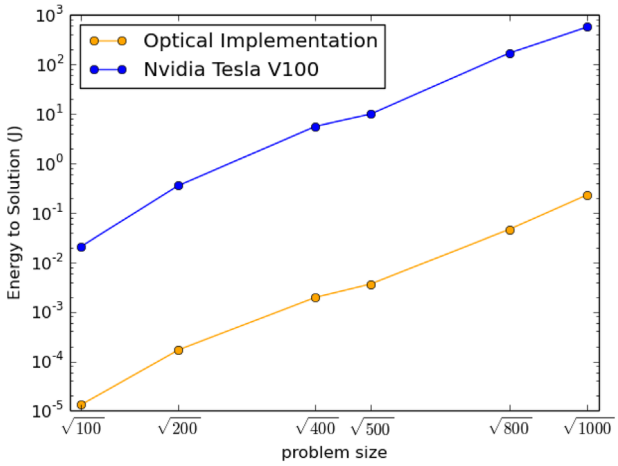

Figure 9. Estimated energy cost to solution of optical and GPU implementations of CIM-CAC versus problem size $\sqrt{N}$. The energy cost to solution for the optical CIM is based on the results listed in Table 2. The energy cost of the GPU is based on the $\tilde{2}00$ W power consumption of the Nvidia Tesla V100 GPU used.

Table 1. Operational power of active photonic devices in 100 GHz CIM.

Devices	Power consumption	Reference
Soliton frequency comb generator	100 mW	[16]
Phase sensitive amplifier (PSA) 10 dB gain	10 mW	[15]
Phase sensitive amplifier (PSA) 50 dB gain	100 mW	[15]
EOM modulator	400 mW	[17]

for PSA_p is ≈1 pJ, whereas the cost of each of the N PSA_s used in the storage ring cavities is ≈100 fJ. Note that the N EOM_s (EOM_1, EOM_2, … , EOM_N) need only be operated for one initial round-trip time, $10^{-11}N$ (s). The operational powers of the active devices in the 100-GHz CIM are summarized in **Table 1**. The energy cost in the pump pulse factory is $E_{\text{factory}} = [1.3 \times 10^{-11} N(\text{MVM}) + 4 \times 10^{-12} N^2 + (10^{-12} + 10^{-13} N)(\text{MVM}) N](J)$. **Table 2** summarizes the energy costs in the three parts of the CIM.

Figure 9 shows the energy cost to solution if the CIM-CFC algorithm is implemented on a graphics processing unit (GPU). A detailed description of this approach will be given in the next section. Even though the optical implementation of the error correction circuit and pump pulse factory as described in Figure 6 is technologically challenging, the energy cost can be decreased by several orders of magnitude relative to a modern GPU.

Table 2. Energy cost to solution in three subsystems in CIM. MVM: matrix-vector multiplication steps to solution, N: problem size, one round trip time: 10^{-8} s, signal lifetime: 10^{-7} s.

Subsystem	Energy-to-solution
Main cavity	$2.6 \times 10^{-20}(\text{MVM})\ \text{N g}^{-2}$
Optical error correction circuit	$[(N+1)10^{-13} + 10^{-12}](\text{MVM})N \simeq 10^{-13}N^2(\text{MVM})$
Pump pulse factory	$[N \times 10^{-13} + 10^{-12}](\text{MVM})N + 1.3 \times 10^{-11}N(\text{MVM}) + 4 \times 10^{-12}N^2 \simeq 10^{-13}N^2(\text{MVM})$

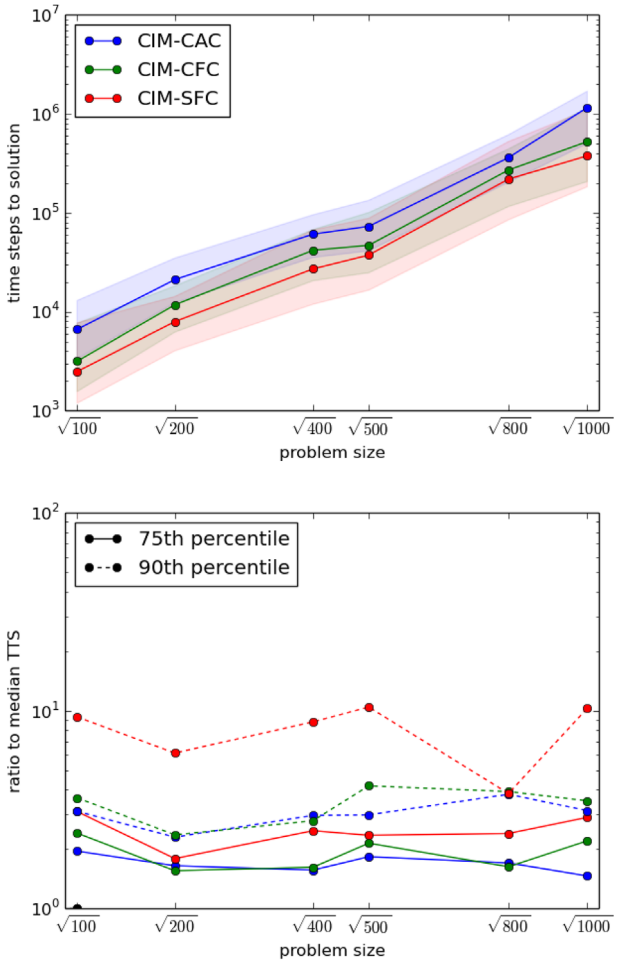

Figure 10. (top) TTSs of CIM-CAC, CIM-CFC, and CIM-SFC vs. problem size $\sqrt{N}$. The shaded regions represent the 25th–75th-percentile TTS. (bottom) The 75th and 90th-percentile TTS compared with the median TTS as a function of problem size for all three systems.

6. CIM-Inspired Heuristic Algorithms

6.1. Scaling Performance of CIM-CAC, CIM-CFC, and CIM-SFC

To test whether the three classical nonlinear dynamics models given by Equations (1)–(8) are good Ising solvers, we can numerically integrate them on a digital platform. In this section, we will consider the numerical integration of these CIM-inspired algorithms using an Euler step. To ensure numerical stability, we constrain the range of some variables; the details of this are presented in Appendix A.

The relevant performance metric is the time to solution, or TTS (the number of integration time steps required to achieve a success rate of 99%). In particular, we study how the median TTS scales as a function of the problem size for randomly generated SK spin glass instances (the couplings are chosen randomly from the range +1 to −1). The median TTS is computed on the basis of a set of 100 randomly generated instances per problem size; 3200 trajectories are used per instance to evaluate the TTS.

Figure 10 shows the median TTSs of the three CIM-inspired algorithms (CIM-CAC, CIM-CFC, and CIM-SFC) with respect

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to problem size. The shaded regions represent 25th–75th percentiles. The linear behavior of the TTS with respect to $\sqrt{N}$ indicates that these algorithms have the same root exponential scaling of TTS that is observed in physical CIMs with quantum noise from external reservoirs.[8,31] All three algorithms appear to have very similar scaling coefficients if the TTS is assumed to be of the form TTS $\approx A \cdot B^{\sqrt{n}}$. In addition to the similar scaling, all three algorithms show a similar spread (25th–75th percentile) in TTS, as indicated by the shaded region in the top figure. Although CIM-SFC appears to have a slightly larger spread in all cases, its spread does not appear to increase at larger problem sizes.

6.2. Comparison with Noisy Mean Field Annealing

To demonstrate the importance of the auxiliary variable (error pulse) in CIM-SFC, we compared its performance to that of another CIM-inspired algorithm—noisy mean field annealing (NMFA).[26] Although NMFA also applies a hyperbolic tangent function to the mutual coupling term, it does not have an auxiliary variable and relies on (artificial) quantum noise to escape from local minima. **Figure 11** compares the scaling of NMFA to CIM-SFC with different values of the feedback parameter k. As k controls the strength of the destabilization force caused by the auxiliary variable, we can measure the importance of the term $k(z_i - e_i)$ to the scaling behavior. When $k = 0$, CIM-SFC is nearly identical to NMFA. The fact that CIM-SFC performs slightly worse at $k = 0$ indicates that the noise included in NMFA likely has a small effect and might help destabilize the local minima (which can also be observed in Figure 7). The cases $k = 0.15$ and $k = 0.2$ are shown as, respectively, an intermediate case and as the (experimentally obtained) optimal value for k in CIM-SFC.

It is seen that the addition of the error correction feedback term $k(z_i - e_i)$ to Equation (7) is effective at improving both the scaling and spread of TTS for the SK instances. This implies that the "correlated artificial noise" provided by the auxiliary variable is more effective at finding better solutions than the "random quantum noise" from the reservoirs.

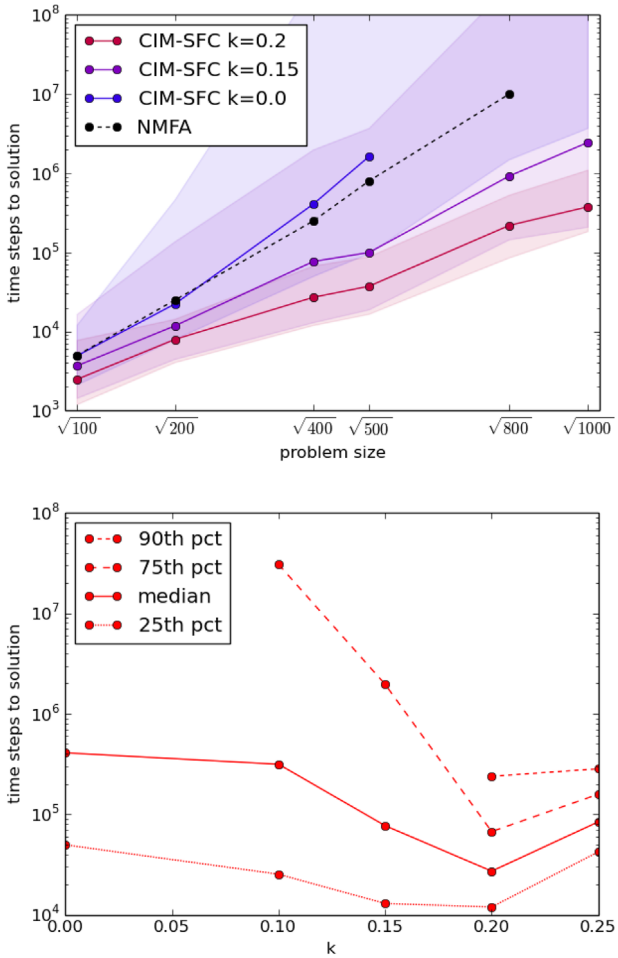

Figure 11. (top) TTSs of CIM-SFC and NMFA versus problem size $\sqrt{N}$. The shaded regions represent 25th–75th-percentile TTS. The results for NMFA are from ref. [12]. (bottom) TTS for $N = 400$ SK instances for different values of k. The 75th and 90th percentiles are not shown for smaller values of k because they were too large to be computed.

6.3. Comparison with Discrete Simulated Bifurcation Machine

We then compared the performance of the CIM-inspired algorithms with that of another heuristic Ising solver—the discrete simulated bifurcation machine (dSBM).[27–29] Like CIM, dSBM makes use of analog spins and continuous dynamics to solve combinatorial optimization problems.

We are aware that the authors of ref. [29] seem to claim that dSBM is algorithmically superior to CIM-CAC based on their comparison of the required number of MVMs to solution. Although the authors of ref. [29] discussed the wall clock TTS of their implementations on many problem sets, in making the claim of algorithmic superiority they only used the median TTS (in units of MVM) on SK instances with two problem sizes. In this section, we will provide a more detailed comparison between the three algorithms (CIM-CAC, CIM-CFC, and CIM-SFC) and dSBM using MVM to solution (or equivalently, the number of integration time steps to solution) as the performance metric. As mentioned before, this provides for effective comparison because all of the algorithms in question will have MVM as a computational bottleneck when implemented on a digital platform. As discussed in Section 4, the computation of the Ising energy can be left until the end of the trajectory in most cases; thus, for this section we will only consider the MVM involved in the computation of the mutual coupling term when calculating the MVM to solution.

The following problem instance sets were used in the comparison:

1) A set of 100 randomly generated 800-spin SK instances (available upon request from the authors). This instance set contains fully connected instances with weights of +1, −1.
2) The G-set instances that have been used as a benchmark for max-cut performance (available at https://web.stanford.edu/ yyye/yyye/Gset/). In this study, we considered 50 instances with problem sizes of 800–2000. These instances have

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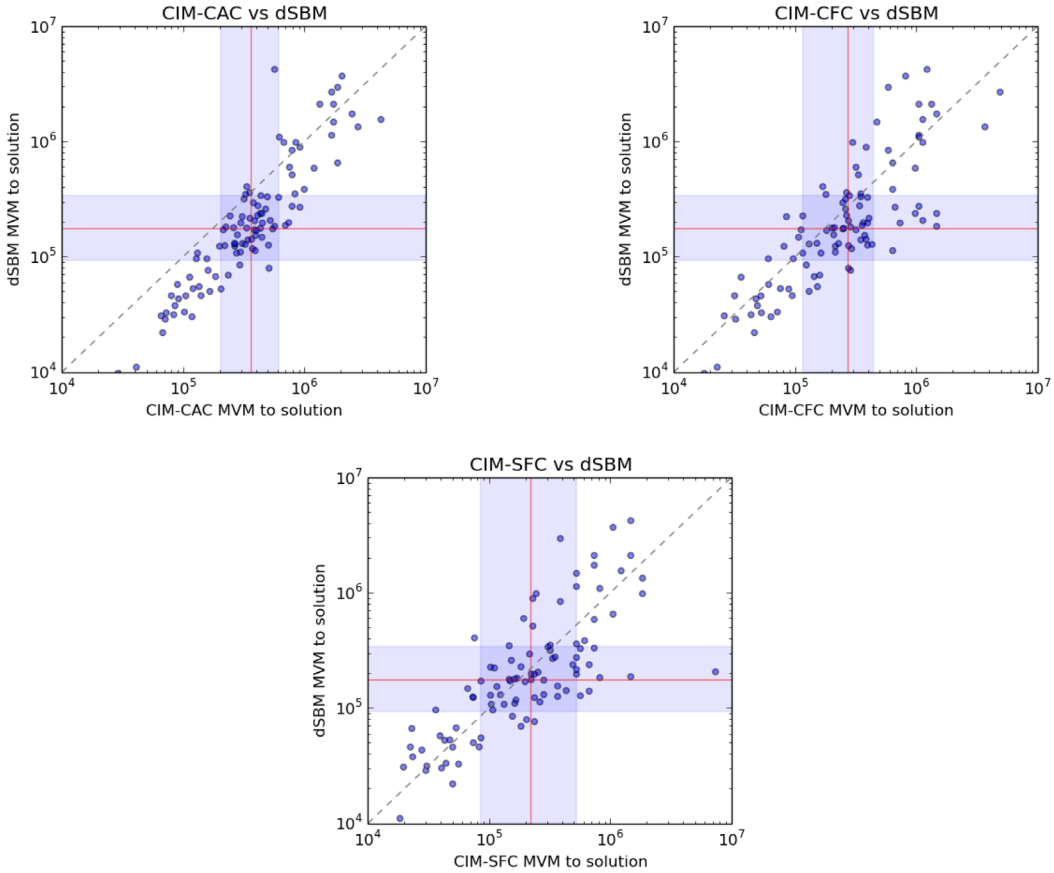

Figure 12. Required number of MVMs to solution for CIM-CAC, CIM-CFC, and CIM-SFC versus number of MVMs to solution for dSBM. The median TTSs are indicated by red lines and the 25th–75th percentiles are indicated by shaded blue regions.

varying edge densities and include weights of either $+1, 0$ or $+1, 0, -1$.

3) Another set of 1000 randomly generated 800- and 1200-spin SK instances (available upon request from the authors) was used to evaluate the worst-case performance.

To compare the performance on the 800-spin SK instances, the dSBM algorithm was also implemented on a GPU. The parameters for dSBM were chosen on the basis of the parameters in ref. [29] (see Appendix D).

Figure 12 compares the performance of the three algorithms (CIM-CAC, CIM-CFC, and CIM-SFC) in each instance with that of dSBM on the 800-spin SK instance set. The ground-state energies used to evaluate the MVM to solution are the lowest energies found by the four algorithms. As all four algorithms found the same lowest energies, it is highly likely that these are true ground-state energies. The parameters for all four systems can be found in Appendix A. As shown in Figure 12, all four systems performed in a remarkably similar manner on the 800-spin-instances case when the parameters were optimized. It is important to note that, using the parameters shown in Figure 12, CIM-SFC did not find the ground state in one instance. However, if different parameters are used, CIM-SFC will find the ground state for this particular instance as well. Thus, although CIM-SFC can achieve high performance, it is highly sensitive to parameter selection.

The median TTSs (in units of MVM) of CIM-CFC, CIM-SFC, and dSBM were nearly the same at $\approx 2 \times 10^5$ each. Furthermore, the spread in TTS was similar for all three algorithms. Although CIM-CAC had a slightly worse median TTS (by less than a factor of two), it is worth noting that the instances in which CIM-CAC performed better than dSBM tended to be the harder instances, indicating that, of the four algorithms, CIM-CAC might show slightly better worst-case performance. We investigate this further later in this section. This pattern was also seen on the *G*-set.

Overall, all four algorithms performed similarly on the fully connected instance set, and it was impossible to determine which algorithm was the most effective for this problem type. In addition to the similarities in terms of median TTS and spread, there were high levels of correlation between the TTSs produced by the four systems. This indicates either that instance difficulty is a universal property for all Ising heuristics or that there is something fundamentally common to the four algorithms: see Appendix E for a further discussion of the similarities and differences between the systems.

Although CIM-SFC performed well on fully connected problem instances, it struggled on many *G*-set instances. In Appendix D, we discuss a partial reason for this failure; however,

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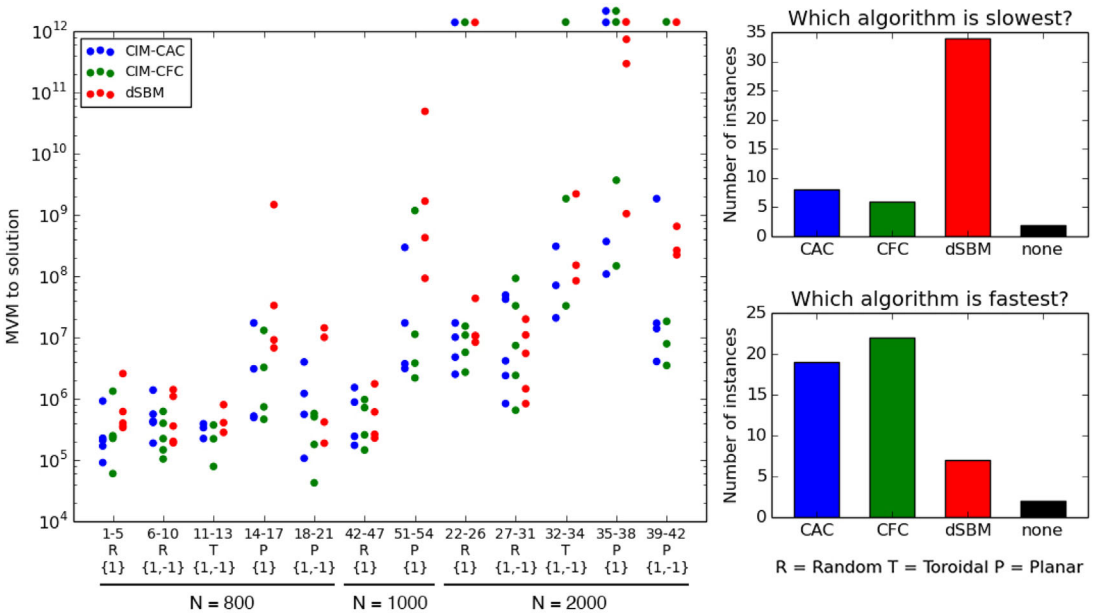

Figure 13. (left) TTS on *G*-set graphs for CIM-CAC, CIM-CFC, and dSBM. The best known cut values are found in ref. [29]. The TTS for dSBM is from ref. [29]. In this plot, the instances are separated into groups depending on the graph type and size. The dots above the plot indicate where the best known cut value was not found. (right) Histograms showing which algorithm achieves the slowest and fastest TTS, respectively. The column labeled "none" indicates the two instances in which none of the three algorithms found the best known cut value. The parameters are chosen and optimized separately for each instance type. An instance-by-instance comparison and the parameters used are presented in Appendix B.

the full reason is yet to be understood. In the future, we expect to modify CIM-SFC or find better parameters to enable it to solve all problem types; for now, we will simply consider CIM-SFC to be a fully connected (or densely connected) Ising solver and compare only the performances of CIM-CAC, CIM-CFC, and dSBM on the *G*-set. For some of the results produced by CIM- SFC on the *G*-set, see Appendix D.

The three algorithms (CIM-CAC, CIM-CFC, and dSBM) all demonstrated fairly good performance on the *G*-set; however, we argue that CIM-CAC is the most consistently effective algorithm. CIM-CAC and dSBM were able to find the best known cut values in 47 out of 50 instances; CIM-CFC found the best known cut value in 45 out of 50 instances. It is worth noting that the simulation time used to calculate the TTS for dSBM[29] was much longer than that used in this study. Given the same simulation time, dSBM would most likely have solved only 45 out of 50 instances. As shown in **Figure 13**, CIM-CAC and CIM-CFC were faster (in units of MVM) than dSBM in most instances. More importantly, among the instances in which dSBM was faster there were no cases in which it significantly faster than CIM-CAC other than G37, in which CIM-CAC did not find the best known cut value. Meanwhile, CIM-CAC was more than an order of magnitude faster than dSBM in 13 out 50 instances. Based on this, we believe that CIM-CAC is a more reliable algorithm when considering many problem types.

The difference between CIM-CAC and CIM-CFC is subtle, which is to be expected given that the dynamics of the two systems are very similar. Although the performance of the two algorithms on the *G*-set was nearly identical in most cases, for some of the harder instances there were some cases in which CIM-CFC could not find the best known cut value or CIM-CFC required a significantly longer TTS. This indicates either that CIM- CAC is fundamentally a more promising algorithm or a precise parameter selection for CIM-CFC is required.

As noted in Figure 12, the worst-case performance of CIM-CAC might have been slightly better than that of dSBM. To evaluate this further, we created new sets of 1000 800- and 1200-spin SK instances. **Figure 14** shows the number of instances solved as a function of the number of MVMs required to achieve a success probability of 99%. It is seen that, in both cases, dSBM could solve the easier instances with fewer MVMs but, for the hardest instances, CIM-CAC was faster; this can be ascertained by observing the intersection point of the two curves in Figure 14.

In nearly all cases, the best Ising energy was the same for both solvers when a similar number of MVMs was used (see Appendix A for the parameters). However, for two instances in the $N = 1200$ set, the Ising energy found by CAC was not found by dSBM. This remained true even when 50 000 dSBM trajectories were used for these instances.

Our results suggest that dSBM might struggle considerably on some harder SK instances. However, we acknowledge that this could be a result of sub-optimal parameter selection for dSBM. The parameters used (see Appendix A) were optimized manually to achieve a good median TTS; however, they might not be the best parameters for solving the hardest instances. By contrast, for CIM-CAC the optimal parameters for the median TTS appear to also perform well on the hardest instances.

To ensure that an Ising solver can find the true ground state of a given problem, it is very important that its worst-case performance is acceptable. For this purpose, we believe that CIM-CAC is likely to be the most fundamentally superior algorithm, at least

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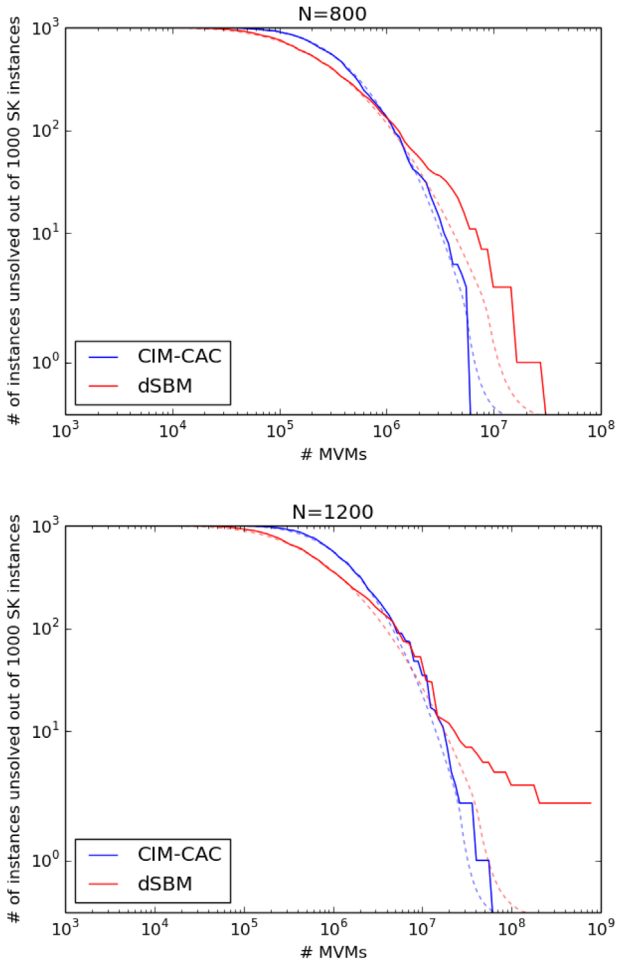

Figure 14. Number of instances that remain unsolved (success probability under 99%) after a certain number of MVMs (time steps) for CIM-CAC and dSBM. The dotted lines represent the assumption of a log-normal distribution for the TTS on randomly generated SK instances. The instance sets used are 1000 randomly generated SK instances (different from the 100 instances used in Figure 12) of problem sizes $N = 800$ (top) and $N = 1200$ (bottom). The parameters for both systems can be found in Appendix A.

in the case of randomly generated SK instances. For the other CIM modifications, this is likely not true; in particular, for CIM-SFC the worst-case performance is significantly worse than that of dSBM and CIM-CAC (as shown in Figure 10). In the future, it would be interesting to investigate the cause of this phenomenon and also to examine the spread of TTS values over different problem types.

7. Conclusion

The new coherent Ising machines presented in this paper (CIM-CFC and CIM-SFC) have considerable potential as both digital heuristic algorithms and optically implemented physical devices. Rapid advances in thin-film $LiNbO_3$ platform[15–17] as a photonic integrated circuit technology might enable the proposed optical CIM to surpass existing digital algorithms implemented on CMOS platforms in terms of both speed and energy consumption.

The proposed CIM-inspired algorithms were shown to be fast and accurate Ising solvers even when implemented on an existing digital platform. In particular, we demonstrated that their performance is very similar to that of other existing analog-system-based algorithms such as dSBM. This again brings up the question raised in ref. [11] as to whether the simulation of analog spins on a digital computer can outperform a purely discrete heuristic algorithm. Finally, whether chaotic dynamics are necessary for a deterministic dynamical system to be a good Ising solver is left as an open issue for future research.[11,12,29]

Appendix A: Optimization of Simulation Parameters

Here we summarize the simulation parameters used in our numerical experiments. The parameters are optimized empirically and, therefore, do not necessarily reflect the true optimum values.

Parameters used in Figure 5

CIM-SFC (upper left panel)		CIM-CFC (upper right panel)	
N step	500	N step	1000
ΔT	0.4	ΔT	0.4
p	−1.0	p	−1.0
c	1.0	α	1.0
β	0.3	β	0.2
k	0.2		
CIM-SFC (lower left panel)		**CIM-CFC (lower right panel)**	
N step	500	N step	1000
ΔT	0.4	ΔT	0.4
p	$-1.0 \rightarrow 1.0$	T_r	900
c	$1.0 \rightarrow 3.0$	T_p	100
β	$0.3 \rightarrow 0.1$	p	$-1.0 \rightarrow 1.0$
k	0.2	α	1.0
		β	0.2

Parameters used in Figures 10, 11, and 12

CIM-CAC: In the simulation of CIM-CAC, the x_i variables were restricted to the range $[-\frac{3}{2}\sqrt{\alpha}, \frac{3}{2}\sqrt{\alpha}]$ at each time step. The parameters p and α were modulated linearly from their starting to ending values during the T_r time steps and were kept at the final value for an additional T_p time steps. The initial value x_i was set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 10^{-4} and $e_i = 1$. Furthermore, 3200 trajectories were computed per instance to evaluate TTS. The actual parameters used for simulation are listed below:

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N step	3200
ΔT	0.125
T_r	2880
T_p	320
p	$-1.0 \rightarrow 1.0$
α	$1.0 \rightarrow 2.5$
β	0.8

CIM-CFC: In the simulation of CIM-CFC, the x_i variables were restricted to the range $[-1.5, 1.5]$ and e_i was restricted to the range $[0.01, \infty]$. The parameter p was modulated linearly from its starting to ending values during the first T_r time steps and kept at the final value for an additional T_p time steps. The initial value x_i was set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 0.1 and $e_i = 1$. Furthermore, 3200 trajectories were computed per instance to evaluate TTS. The actual parameters used for simulation are listed below:

N step	1000
ΔT	0.4
T_r	900
T_p	100
p	$-1.0 \rightarrow 1.0$
α	1.0
β	0.2

CIM-SFC: The x_i and e_i variables did not need to be restricted as this system is more numerically stable than CIM-CAC or CIM-CFC. The parameters p, c, and β were modulated linearly from their starting to ending values during simulation. The initial value x_i was set to a random value chosen from a zero-mean Gaussian distribution with standard deviation of 0.1 and $e_i = 0$. 3200 trajectories were computed per instance to evaluate TTS. The actual parameters used for simulation are listed below:

N step	500
ΔT	0.4
p	$-1.0 \rightarrow 1.0$
c	$1.0 \rightarrow 3.0$
β	$0.3 \rightarrow 0.1$
k	0.2

In addition to the parameters mentioned above, it is important that the normalizing factor ξ for the mutual coupling term be chosen as[31]

$$\xi = \sqrt{\frac{2N}{\sum J_{ij}^2}} \tag{A1}$$

This choice is crucial for the successful performance of CIM-SFC but not for CIM-CAC or CIM-CFC.

Moreover, it is important to note that we used the same number of time steps for all problem sizes in Figures 10 and 11. As it is likely that the optimal number of time steps is smaller for smaller problem sizes, the scaling of TTS when the number of time steps is optimized separately for each problem size might be slightly worse than the reported scaling. However, we do not believe that this difference would be very significant. For a discussion of the scaling of TTS for CIM-CAC when different parameters are chosen (see Appendix C).

dSBM: To obtain the results shown in Figure 12, dSBM was implemented as described in ref. [29]. The following parameters were used:

N step	2000
ΔT	1.25
c	0.5

Parameters used in Figure 14

The parameters for the $N = 800$ case were the same as those used to obtain Figure 12. The parameters used for $N = 1200$ are listed below. In most instances, 3200 trajectories were used for $N = 1{,}200$; however, to accurately evaluate the probability of success, 10 000–50 000 trajectories were computed for the ten hardest instances for both algorithms. Moreover, owing to the hardness in the $N = 1200$ case, we are not very certain that the true ground state was found.

CIM-CAC:

N step	8000
ΔT	0.125
T_r	7200
T_p	800
p	$-1.0 \rightarrow 1.0$
α	$1.0 \rightarrow 2.5$
β	0.8

dSBM:

N step	4000
ΔT	1.25
c	0.5

Numerical Integration

An Euler step was used for integration in all cases except for dSBM. As described above, we constrained the ranges of the x_i variables to ensure numerical stability. This was not necessary to ensure performance, but it allowed us to increase the integration time step by a factor of two or three without compromising the probability 0f success. In **Figure A1**, we show the probability of success of CIM-CAC with respect to the time step value for both constrained and unconstrained systems.

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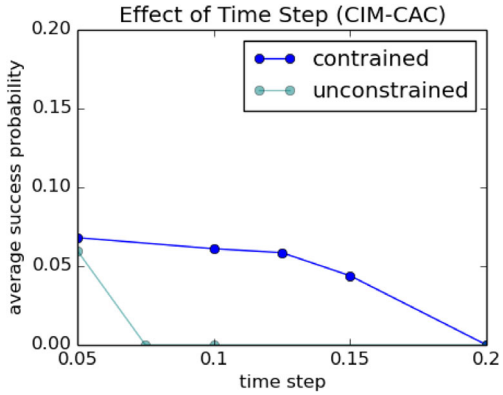

Figure A1. Probability of success of CIM-CAC with respect to time step for both constrained and unconstrained systems. For the blue curve, the x_i amplitudes are restricted to the range $[-1.5\sqrt{\alpha}, 1.5\sqrt{\alpha}]$.

The results in Section 5 for CIM-CFC do not reflect the use of this numerical constraint; the CIM in this section was meant to be a physical machine and, therefore, a time step of 0.2 was used.

Appendix B: Simulation Results for *G*-Set

The results in Figure 13 for dSBM are taken directly from the GPU implementation of dSBM in ref. [29]. The unit for TTS in **Table B1** is TTS or, equivalently, MVM to solution. In our simulation, 3200, 10 000, or 32 000 trajectories were generated to evaluate the TTS depending on the instance difficulty. The numbers in bold denote the best TTS among the three algorithms.

CIM-CAC Parameters for G-Set

The variables were restricted as described in Appendix A and the initial conditions were set in the same manner. The following parameters were the same for all the *G*-set instances:

$$\begin{array}{c|c} \alpha & 1.0 \rightarrow 3.0 \\ \hline \beta & 0.3 \end{array} \tag{B1}$$

The parameters p, ΔT, and the number of time steps used in each phase were chosen by instance type as follows:

Graph type	Edge weight	N	Instance #	p	N step	ΔT	T_r	T_p
Random	{+1}	800	1–5	−0.5 → 1.0	6666	0.075	6000	666
Random	{+1, −1}	800	6–10	−0.5 → 1.0	6666	0.075	6000	666
Toroidal	{+1, −1}	800	11–13	−4.0	5000	0.1	4500	500
Planar	{+1}	800	14–17	−1.0	20 000	0.05	18 000	2000
Planar	{+1, −1}	800	18–21	−1.0	20 000	0.05	18 000	2000
Random	{+1}	1000	43–46	−0.5 → 1.0	10 000	0.1	9000	1000
Planar	{+1}	1000	51–54	−1.0	20 000	0.05	18 000	2000
Random	{+1}	2000	22–26	−0.5 → 1.0	20 000	0.1	19 000	1000
Random	{+1, −1}	2000	27–31	−0.5 → 1.0	20 000	0.1	19 000	1000
Toroidal	{+1, −1}	2000	32–34	−4.0 → −3.0	20 000	0.1	19 000	1000
Planar	{+1}	2000	35–38	−1.0 → −0.5	80 000	0.05	78 000	2000
Planar	{+1}	2000	39–42	−1.0 → −0.5	80 000	0.05	78 000	2000

CIM-CFC Parameters for G-Set

The variables were restricted as described in Appendix A and the initial conditions were set in the same manner. The following parameters were the same for all the *G*-set instances:

$$\begin{array}{c|c} \alpha & 1.0 \\ \hline \beta & 0.15 \end{array} \tag{B2}$$

The parameters p, ΔT, and the number of time steps used in each phase were chosen by instance type as follows:

Graph type	Edge weight	N	Instance #	p	N step	ΔT	T_r	T_p
Random	{+1}	800	1–5	−1.0 → 1.0	4000	0.125	3600	400
Random	{+1, −1}	800	6–10	−1.0 → 1.0	2000	0.25	1800	200
Toroidal	{+1, −1}	800	11–13	−3.0 → −1.0	2000	0.25	1800	200
Planar	{+1}	800	14–17	−2.0 → 0.0	8000	0.125	7200	800
Planar	{+1, −1}	800	18–21	−2.0 → 0.0	4000	0.25	3600	400
Random	{+1}	1000	43–46	−1.0 → 1.0	5000	0.2	4500	500
Planar	{+1}	1000	51–54	−2.0 → 0.0	16 000	0.125	15 200	800
Random	{+1}	2000	22–26	−1.0 → 1.0	10 000	0.2	9500	500
Random	{+1, −1}	2000	27–31	−1.0 → 1.0	10 000	0.2	9500	500
Toroidal	{+1, −1}	2000	32–34	−3.0 → −1.0	40 000	0.1	39 000	1000
Planar	{+1}	2000	35–38	−2.0 → 0.0	80 000	0.05	78 000	2000
Planar	{+1}	2000	39–42	−2.0 → 0.0	40 000	0.1	39 000	1000

Appendix C: Reasoning for Parameter Selection

The parameters were generally selected numerically; however, the choices of p, α, and β can be understood as follows. It is observed that the average residual energy visited by CIM-CAC during the search process can be roughly estimated by the formula[11]

$$\Delta E_{\text{avg}} \approx K \frac{1-p}{\alpha\beta} \tag{C1}$$

where K is a constant depending only on the problem type and size. This formula essentially predicts the effective sampling temperature of the system (although the distribution might not be an exact Boltzmann distribution). Based on this philosophy, we gradually reduced the "system temperature" to produce an annealing effect. This was the motivation for increasing p and α. The different choices for the range of p on different *G*-set instances reflect the vastly different values for the constant K depending on the structure of the max-cut problem.

In a more general setting, the value of K can be predicted on the basis of the problem type and the ranges for p and α can be chosen accordingly.

Although it has not been verified, a similar formula most likely holds for CIM-CFC; thus, the parameters for CIM-CFC were chosen in the same manner.

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Table B1. Probabilities of success and TTSs of CIM-CAC, CIM-CFC, and dSBM on *G*-set graphs. The probabilities of success for CIM-CAC and CIM-CFC are for single trajectories with the parameters detailed in Appendix B. Note that P_s for dSBM taken from ref. [29] is the success probability for a batch of 160 trajectories. To calculate the P_s for a single dSBM trajectory, use $1-(1-P_s)^{\frac{1}{160}}$.

Instance	CIM-CAC TTS	CIM-CAC P_s	CIM-CFC TTS	CIM-CFC P_s	dSBM TTS	dSBM P_s
G1	90 805	0.286875	**60 078**	0.264062	339 332	0.987
G2	**920 217**	0.0328125	1 330 454	0.01375	2 578 124	0.82
G3	**169 551**	0.165625	249 212	0.07125	400 343	0.996
G4	**209 086**	0.136563	239 389	0.0740625	361 673	0.983
G5	226 881	0.126562	**226 449**	0.078125	618 221	0.972
G6	188 908	0.15	**104 083**	0.0846875	190 728	0.979
G7	562 413	0.053125	**146 490**	0.0609375	201 889	0.974
G8	431 029	0.06875	399 118	0.0228125	**358 947**	0.954
G9	411 604	0.071875	**223 836**	0.0403125	1 095 704	0.867
G10	1 388 073	0.021875	**622 470**	0.0146875	1 410 031	0.407
G11	337 563	0.0659375	**222 079**	0.040625	282 524	0.98
G12	224 452	0.0975	**78 562**	0.110625	407 997	0.973
G13	391 011	0.0571875	**373 236**	0.024375	800 687	0.996
G14	17 291 018	0.0053125	**13 080 721**	0.0028125	1 469 967 245	0.005
G15	521 572	0.161875	**462 528**	0.0765625	6 782 113	0.804
G16	**494 303**	0.17	737 146	0.04875	9 156 329	0.992
G17	**3 089 154**	0.029375	3 256 332	0.01125	33 222 397	0.283
G18	556 635	0.1525	**507 805**	0.035625	14 375 986	0.074
G19	1 218 307	0.0728125	**179 562**	0.0975	417 204	0.995
G20	106 718	0.578125	**42 428**	0.352187	188 349	0.98
G21	3 991 180	0.0228125	**574 365**	0.0315625	10 080 921	0.136
G43	174 031	0.2325	**145 625**	0.14625	228 908	0.992
G44	**244 188**	0.171875	257 230	0.085625	263 171	0.985
G45	**880 856**	0.0509375	970 877	0.0234375	1 754 473	0.985
G46	1 528 073	0.0296875	717 957	0.0315625	**610 421**	0.992
G51	3 732 360	0.024375	**2 189 016**	0.0331	424 989 992	0.067
G52	**3 122 871**	0.0290625	382 0774	0.0191	92 285 270	0.213
G53	17 291 018	0.0053125	**11 298 922**	0.0065	1 676 440 469	0.043
G54	**294 684 837**	0.0003125	1 178 886 725	6.25e-05	49 107 077 298	0.0006
G22	**2 516 544**	0.0359375	2 718 081	0.0168	8 401 394	0.928
G23	N/A	0.0	N/A	0.0	N/A	0.0
G24	**4 785 454**	0.0190625	5 733 406	0.008	10 585 339	0.648
G25	17 291 018	0.0053125	**15 327 529**	0.003	43 414 254	0.399
G26	**10 117 012**	0.0090625	10 941 648	0.0042	10 730 290	0.643
G27	835 530	0.104375	**649 972**	0.0684	832 465	0.971
G28	2 389 444	0.0378125	2 413 498	0.0189	**1 455 914**	0.952
G29	**4 164 219**	0.021875	7 404 644	0.0062	5 516 820	0.737
G30	42 058 344	0.0021875	32 871 041	0.0014	**11 002 259**	0.738
G31	49 075 749	0.001875	92 080 375	0.0005	**19 923 732**	0.199
G32	**70 802 710**	0.0013	1 841 975 969	0.0001	150 969 342	0.093
G33	**306 965 291**	0.0003	N/A	0.0	2 204 950 868	0.005
G34	**20 886 506**	0.0044	32 801 883	0.0056	84 156 149	0.231
G35	N/A	0.0	N/A	0.0	N/A	0.0
G36	**368 321 503**	0.0005	3 683 951 938	0.0001	736 790 387 782	0.0001
G37	N/A	0.0	N/A	0.0	**294 701 417 831**	0.0002
G38	**108 264 816**	0.0017	147 181 162	0.0025	1 046 295 719	0.068
G39	4 065 368	0.0443	**3 497 864**	0.0513	651 087 484	0.107
G40	1 841 975 969	0.0001	N/A	0.0	**264 354 894**	0.154
G41	17 123 320	0.0107	**7 916 531**	0.023	222 414 431	0.282
G42	**13 969 282**	0.0131	18 328 423	0.01	N/A	0.0

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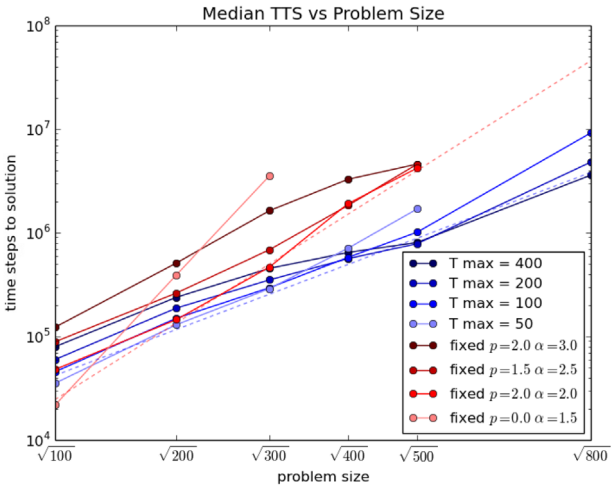

Figure C1. Performance of CIM-CAC with respect to problem size for different parameters. The fixed parameters are indicated in red and the parameter modulation is indicated in blue. The red and blue dotted lines are fits for the lower envelopes of the red and blue curves, respectively.

Optimal Parameters with Respect to Problem Size (CIM-CAC)

Figure C1 shows the difference in scaling when the parameters are fixed (red-shaded areas) compared to when the parameters are modulated linearly (blue-shaded areas). In addition, the optimal annealing time (in other words, the optimal speed of modulation) will change with respect to the problem size, as long annealing times are required to obtain good results at large problem sizes. This pattern was also used in choosing the parameters on the *G*-set.

Although Figure C1 is based on the Gaussian quantum model for CIM-CAC in MFB-CIM (see ref. [31]), the difference in performance between this model and the noiseless model discussed in this paper is insignificant, as $g^2 = 10^{-4}$ was used. It should also be noted that a time step of 0.01 was used to produce the results in Figure C1; as a result, the TTS in this figure is an order of magnitude longer than the results produced in this study, in which a time step of 0.125 was used.

Appendix D: Results and Discussion for CIM-SFC on G-Set

Based on our understanding of CIM-SFC, it is very important that the term $\tanh(cz_i)$ transitions from the "soft spin" mode, in which $cz_i \approx 0$ and $\tanh(cz_i) \approx cz_i$, to the "discrete spin" mode, in which $|cz_i| >> 0$ and $\tanh(cz_i) \approx \text{sign}(cz_i)$. Therefore, we used the normalizing factor ξ (as defined above) to ensure that z_i would on average be around $\sqrt{2}$ for a randomly chosen spin configuration, thereby allowing us to obtain similar results using the same value for c in all of the cases. However, this only works for conditions such as those for SK instances in which each node has equal connectivity; thus, we could expect z_i to have roughly the same range of values for all i.

On some *G*-set instances, in particular the planar graph instances, some nodes had a much larger degree; as a result, cz_i was too large in some cases and too small in others regardless of the normalizing factor ξ used. This might have been one of the reasons why CIM-SFC struggled on many *G*-set instances, particularly planar graphs. It could also be the reason why dSBM struggled on planar graphs as well, as dSBM relies on the same normalizing factor to obtain good results. CIM-CAC and CIM-CFC did not need this normalizing factor, as they automatically compensate for different values of $\sum_j J_{ij}\sigma_j$; this might have been why they performed well on planar graphs.

For toroidal graphs, the opposite is true, as $\sum_j J_{ij}\sigma_j$ can only take on five different values for such graphs. This could mean that the transition from "soft spin" to "discrete spin" is rapid in the case of CIM-SFC, indicating the need to carefully tune the parameters to obtain good results on toroidal graphs.

Although this observation regarding the analog/discrete transition might partially explain the poor results on the *G*-set, it is not a complete explanation. For example, CIM-SFC struggled on some random graphs (such as G9) that do not have the above-mentioned property because each node has similar connectivity.

The results for CIM-SFC on the *G*-set as well as the parameters used are listed below (not all instances were tested):

Results for CIM-SFC on the G-Set

Instance	TTS	P_s
G1	28 470	0.194
G2	1 531 984	0.004
G3	130 388	0.046
G4	435 510	0.014
G5	380 685	0.016
G6	112 834	0.0202
G7	163 316	0.014
G8	125 360	0.0182
G9	4 604 018	0.0005
G10	395 845	0.0058
G11	195 461	0.0572
G12	128 184	0.0859
G13	311 368	0.0363
G43	1 702 012	0.0134375
G44	1 742 812	0.013125
G45	73 671 209	0.0003125
G46	1 927 483	0.011875

For instance, on G14–G21 (800-node planar graphs) and G51–G54 (1,000-node planar graphs), CIM-SFC had a success probability of either zero or a very small nonzero value. Note that 2000-node instances were not tested.

Parameters for CIM-SFC on G-Set

Common Parameters:

$$p| -1.0 \rightarrow 1.0 \tag{D1}$$

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Parameters Selected by Problem Type:

Graph type	Edge weight	N	Instance #	c	β	k	N step	ΔT
Random	{+1}	800	1–5	1.0 → 3.0	0.3 → 0.0	0.2	2666	0.15
Random	{+1, −1}	800	6–10	1.0 → 3.0	0.3 → 0.0	0.2	500	0.4
Toroidal	{+1, −1}	800	11–13	1.4	0.05 → 0.0	0.32	2500	0.4
Random	{+1}	1000	43–46	1.4 → 4.2	0.2 → 0.0	0.2	5000	0.2

The parameters for CIM-SFC are chosen experimentally and the understanding of how the parameters affect the performance and dynamics of the algorithm is therefore limited. Once this system is studied more thoroughly, we will propose a more systematic method of choosing parameters to ensure good performance on many different problem types.

Appendix E: Similarities and Differences Between CIM and SBM Algorithms

Using continuous analog dynamics to solve discrete optimization problems is a somewhat new concept, and it is interesting to compare the different approaches.[11,13,29] In this appendix, we will briefly discuss some similarities and differences among the three CIM-inspired algorithms and the SBM algorithms.

All four systems discussed in Section 6, namely CIM-CAC, CIM-CAC, CIM-SFC, and dSBM, were originally inspired by the same fundamental principle[10,27]:

The function

$$H(x) = \sum_i \left(\frac{x_i^2}{4} - \frac{1-p}{2} \right) x_i^2 + c \sum_i \sum_j J_{ij} x_i x_j \tag{E1}$$

can be used as a continuous approximation of the Ising cost function.

In the original CIM algorithm, gradient descent is used to find the local minima of H as it is deformed by increasing p. This system has two significant drawbacks[10]:

1) Local minima are stable;
2) Incorrect mapping of the Ising problem to the cost function occurs as a result of amplitude heterogeneity.

All four algorithms discussed in Section 6 can be regarded as modifications of the original CIM algorithm that attempt to overcome these two flaws.[11,27–29] In each algorithm, the first flaw is addressed by adding new degrees of freedom to the system, resulting in $2N$ (instead of only N) analog variables for N spins. In SBM, this is done by including both a position vector, x_i, and a velocity/momentum vector, y_i; in the modified CIM algorithms, by contrast, the auxiliary variable e_i is added.

To address the second flaw, the creators of dSBM added discretization and "inelastic walls," whereas in CIM-CFC and CIM-SFC this discretization is not necessary. Using different mechanisms, all three algorithms ensure that the system only has fixed points at the local minima of the Ising Hamiltonian (during the end of the trajectory), which is not true for the original CIM algorithm. Because these systems are fundamentally very similar, it should not be surprising that they achieve similar performance.

We also note that, for dSBM to achieve good performance, it is necessary to use discretization and inelastic walls, which make the system discontinuous. This is particularly useful for implementation on a digital platform, in which discrete processes are preferred; however, when implementing these algorithms on an analog physical platform, it is not preferred. In the cases of CIM-CAC, CIM-CFC, and CIM-SFC, the system evolves continuously, making them much more suitable for analog implementation as exemplified by the optical CIM architecture proposed in this paper.

An interesting difference between the CIM and the original bifurcation machine,[27] which was referred to as aSBM in ref. [29], is that aSBM is a completely unitary dissipation-less system. Because of this, aSBM relies on adiabatic evolution for computation (similar to quantum annealing); the dissipative CIM and other Ising heuristics (such as simulated annealing or breakout local search[14]) , by contrast, rely on some sort of dissipative relaxation. However, the new SBM algorithms in ref. [29] deviate from the concept of adiabatic evolution by adding inelastic walls, making the new bifurcation machine a dissipative system in which information is lost over time. It would be interesting to attempt to understand whether dissipation is in fact necessary for a system to achieve the high performance of the algorithms discussed in this paper. For example, one could modify aSBM in a different manner that addresses the problem of amplitude heterogeneity but retains the adiabatic nature. Whether this is possible is beyond the scope of this paper.

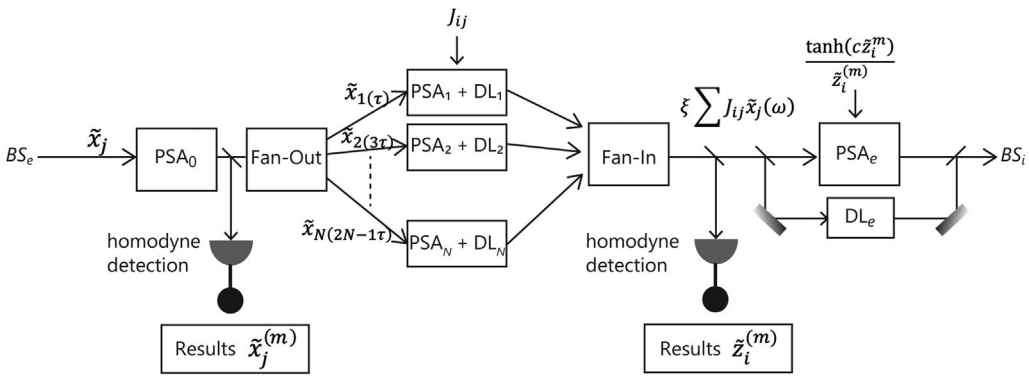

Figure F1. Optical implementation of CIM-SFC.

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Appendix F: Optical Implementation of CIM-SFC

Figure F1 shows an optical implementation of CIM-SFC, which is similar to that of CIM-CAC and CIM-CFC shown in Figure 6. The feedback signal $\tilde{z}_i = \sum_j J_{ij}\tilde{x}_j$ is deamplified (rather than amplified) by $\mathrm{PSA_e}$ with attenuation coefficient $\frac{\tanh(\tilde{z}_i^{(m)})}{\tilde{z}_i^{(m)}}$, where $\tilde{z}_i^{(m)}$ is an optical homodyne measurement result of $\tilde{z}_i$. This feedback signal is then injected back into signal pulse x_i in the main cavity through BS_i.

Part of the fan-in circuit output $\tilde{z}_i$ is delayed by a delay line DL_e with delay time $N\tau$ and combined with the error pulse e_i (inside the main cavity). This implements the term $e_i - \tilde{z}_i$ in Equation (8). The term $-\beta(e_i - \tilde{z}_i)$ on the right-hand side of Equation (8) is implemented by a phase-sensitive amplifier $\mathrm{PSA_e}$ in the main cavity. This is also a deamplification process. Finally, the error correction signal amplitude $e_i - \tilde{z}_i$ is coupled to the signal pulse x_i inside the main cavity using a standard optical delay line.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

coherent Ising machine, chaotic solution search, combinatorial optimization, matrix-vector multiplication, optical error correction

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